

1 Explainability in Reinforcement Learning

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Chapter 1

Deep Learning

This chapter introduces the fundamental concepts of Deep Learning (DL), the training paradigm used to optimize deep Neural Network (NN). The objective is to provide the necessary foundation in DL for the ideas developed throughout the rest of this manuscript. Its structure is as follows:

Section 1.1: An overview of the historical developments that led to the emergence of the field.

Section 1.2: A formalization of the objective underlying DL algorithms.

Section 1.3: An examination of the structure of artificial NN.

Section 1.4: An explanation of how the learning process shapes NN.

Section 1.5: A discussion of how such models are evaluated.

1.1 Historical Developments

DL and NN emerged from an ambition to study the expressive capabilities of organic brain. They can be seen as a particularly striking example of biomimicry, as they draw inspiration from living systems to address scientific challenges. The first major milestone came in 1943 with the proposal of a mathematical model of an artificial neuron, with a formulation of networked interactions [McCulloch and Pitts, 1943]. This work established the key idea that networks of simple units can give rise to complex computations. The focus was on representational power, not on the ability of such systems to learn. As a result, this work remained largely theoretical and had limited practical applications.

A major step toward operationalizing the idea of NNs was achieved with the perceptron algorithm in 1958 [Rosenblatt, 1958]. It was designed to adjust its parameters from input–output pairs in order to approximate a target function. However, its fundamental limitations were exposed in 1969, showing that it could not learn functions as simple as the exclusive-or (XOR) [Minsky and Papert, 1969]. At the same time, it was demonstrated that stacking multiple perceptrons could in principle represent such functions. At the time, no effective training procedure existed for such multi-layer architectures, and it was not until 1986 that researchers proposed using gradient descent optimization to train deep NNs

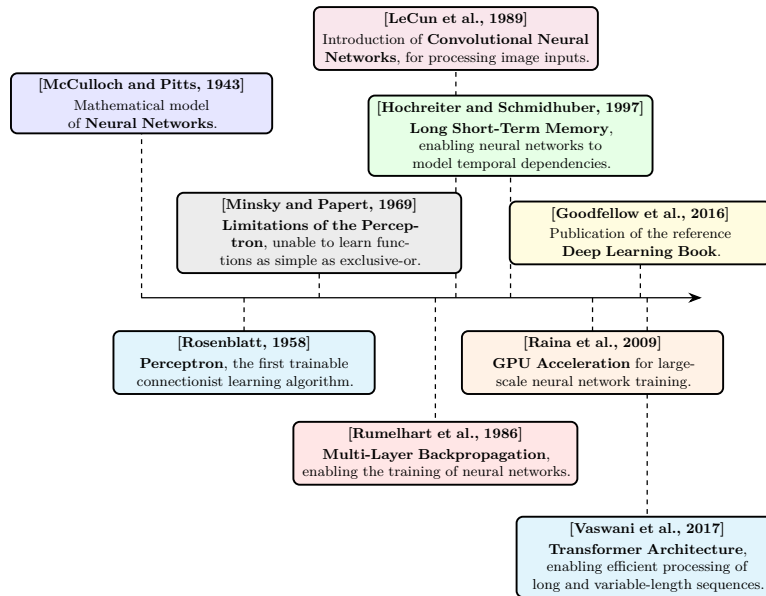


Figure 1.1: Evolution of deep learning.

[Rumelhart et al., 1986]. From this point onward, the emergence of DL is retrospectively situated, even though the term itself appeared later. DL is defined as the training of deep NNs using gradient-based optimization methods. For simplicity, the term NN will refer throughout this manuscript to deep NNs.

It took several decades before DL reached the level of prominence it has today. One major obstacle was the absence of theoretical guarantees regarding the convergence of learning algorithms. Although the universal approximation theorem shows that NN can represent a wide class of functions [Hornik et al., 1989], it did not ensure that such representations could be effectively learned through optimization methods like gradient descent. This lack of rigorous guarantees initially reduced interest within the academic community. As a result, early progress relied largely on empirical breakthroughs.

These empirical advances were made possible by two key developments: the availability of large-scale datasets and significant improvements in computational infrastructure. DL methods require vast amounts of data to perform well, and the rise of the internet alongside the digitization of information enabled the creation of such datasets [Hilbert and López, 2011]. At the same time, progress in hardware, particularly the use of graphics processing units, greatly increased computational capacity and made it feasible to train large-scale models [Krizhevsky et al., 2012].

These changes paved the way for the rise of DL as we know it today. DL has achieved remarkable success across a wide range of domains, including computer vision [Krizhevsky et al., 2012], complex games [Silver et al., 2017a], content recommendation [Covington et al., 2016], bio-chemistry [Jumper et al., 2021], as well as the now essential digital assistants based on Large Language Model (LLM) [Brown et al., 2020]. The diversity of these applications explains the enthusiasm for this technology, which continues to redefine the boundaries of

what was once considered automatable.

Figure 1.1 provides a timeline of the key milestones that shaped deep learning.

1.2 Objective

Now that the main developments that led to the rise of DL have been outlined, it is necessary to formalize what is actually being achieved when performing DL. The range of problems that can be addressed with DL is extremely broad. To ensure clarity, the supervised learning setting will be considered, specifically applied to a regression task. This framework serves as a standard reference in the field, as all building blocks of DL applications are built upon this fundamental formalism.

In this setting, learning consists of progressively shaping a NN that serves as an approximation of a target function. At initialization, the network does not yet provide a meaningful representation of this function. Since a NN is a parametric function, its parameters are progressively updated during training so as to make it a better approximation of the target function. It gradually becomes capable of capturing the relationship between inputs and outputs through successive parameter updates during training. For this reason, DL is naturally situated within the framework of machine learning. A standard definition of machine learning is:

“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”
[Mitchell, 1997]

When this formulation is adapted to DL, the following correspondences hold:

- The computer program corresponds to the learning algorithm, which is responsible for adjusting the network’s parameters.
- The task T consists of learning a NN that provides a good approximation of the target function.
- The performance measure P corresponds to a function used to evaluate the quality of this approximation.
- The experience E takes the form of a dataset composed of input–output pairs derived from the function to be approximated.

The mathematical formalism used in this chapter is now introduced in order to remove any ambiguity in the developments that follow. The function to be approximated is denoted by f . As with any function, it defines a mapping from an input to an output: $x \in \mathcal{X}$ denotes an input and $y \in \mathcal{Y}$ the corresponding output. The function f is therefore characterized by an input space $\mathcal{X}$ and an output space $\mathcal{Y}$, and can be formally written as $f : \mathcal{X} \rightarrow \mathcal{Y}$. For simplicity of exposition and notation, both $\mathcal{X}$ and $\mathcal{Y}$ are assumed to be vector spaces throughout, while keeping in mind that this framework can be readily generalized to more settings. Throughout this manuscript, the notation $\hat{\cdot}$ indicates that a given object is an approximation of another. Since the goal of the trained NN is to approximate f ,

105 this approximation is denoted by $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$. Given an input $x \in \mathcal{X}$, the output
 106 of the approximating function $\hat{f}(x)$ is denoted by $\hat{y}$.

107 It is important to emphasize that the target function f is generally not known
 108 explicitly. If it were, there would be no need to learn an approximation, since
 109 the true function would already provide the exact mapping. In practice, only a
 110 finite set of observations generated by f is available, organized in what is called
 111 a dataset. To distinguish between the different samples, an index i is introduced,
 112 allowing input–output pairs of the form $(x^{(i)}, y^{(i)})$ to be defined. This leads to
 113 the following formulation for a dataset $\mathcal{D}$ of size N , where N denotes the number
 114 of available examples:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^N.$$

115 Now that the way the target function f is represented has been made explicit,
 116 one can ask what it means for a function $\hat{f}$ to be a good approximation of this
 117 function. It means that, for a given input $x^{(i)}$, the NN output $\hat{y}^{(i)}$ is close to the
 118 true output $y^{(i)}$. To quantify the quality of this approximation, a function $\mathcal{L}$ is
 119 introduced that measures the discrepancy between predictions and target values.
 120 In DL, this function called the loss function plays a central role as it guides the
 121 learning process. In regression settings, a common choice is the Mean Squared
 122 Error (MSE), defined as:

$$\mathcal{L}(\hat{y}, y) = (\hat{y} - y)^2.$$

123 Thanks to this loss, it is now possible to rigorously define the performance
 124 measure P . This performance measure corresponds to the average loss over the
 125 dataset and can therefore be written as:

$$P = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\hat{f}(x^{(i)}), y^{(i)}).$$

126 This formulation captures the mathematical core of the learning objective.
 127 The goal is to find a NN $\hat{f}$ that minimizes the prediction error over the dataset
 128 $\mathcal{D}$. As can be seen, the setting considered here is very general, where the target
 129 function f may take many different forms. Consequently, the network $\hat{f}$ must
 130 be expressive enough to approximate a wide variety of such functions. This
 131 requirement motivates the choice of NN architectures, which are specifically
 132 designed to represent highly diverse functional mappings.

133 1.3 Neural Network Structure

134 NN are called connectionist models. Rather than modeling information processing
 135 through a set of explicit rules, they rely on a flow of signals distributed within
 136 a computational structure. Their architecture is composed of a large number
 137 of simple computational units, which operate on partial representations and
 138 exchange signals across the network. A complex pattern of weighted connections
 139 between these units enables the transformation and propagation of information.

140 To simplify the presentation throughout this chapter, the focus is placed
 141 on the MultiLayer Perceptron (MLP) architecture. Although more specialized
 142 architectures exist, the MLP forms a fundamental building block of modern DL.
 143 Indeed, most of the essential mechanisms of DL are present in it, making it a
 144 particularly suitable framework for introducing key concepts.

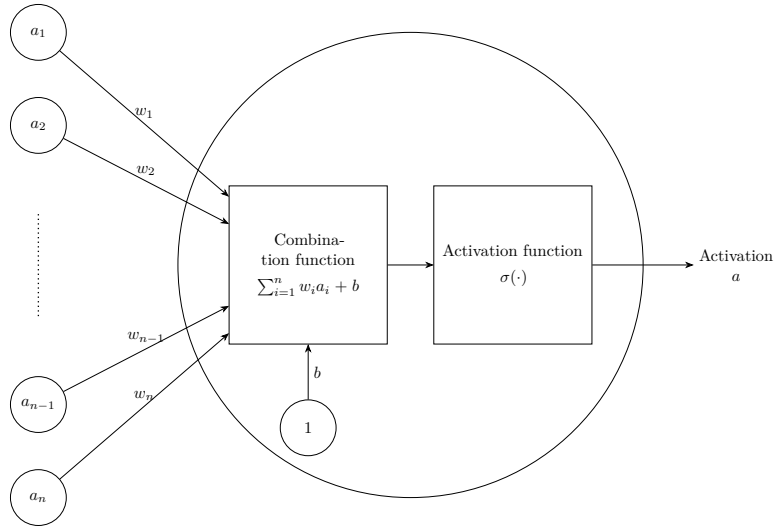


Figure 1.2: Artificiale neurone.

1.3.1 Neurons

The artificial neuron is the basic building block of NN. It is through these units that all computations within the network are carried out. A neuron receives a set of input signals from other neurons, where these inputs are represented as real-valued numbers. It produces an output, also a real-valued number, which is transmitted to subsequent neural units. Its internal state, called the activation, reflects its level of response and encodes the information processed for a given input. This activation, or lack thereof, is then used as input information by other neurons to determine their own activations, thereby propagating information through the network.

A neuron is characterized by:

- a set of incoming connections from other neurons,
- a weight associated with each of these connections,
- a bias term,
- a differentiable nonlinear activation function,
- a scalar activation state.

The activation of a neuron is determined by a weighted combination of its input signals, to which a bias is added, followed by the application of a nonlinear activation function. Mathematically, for a neuron receiving inputs $(a_1, \dots, a_n)$, its activation a is given by:

$$a = \sigma \left(\sum_{i=1}^n w_i a_i + b \right),$$

where:

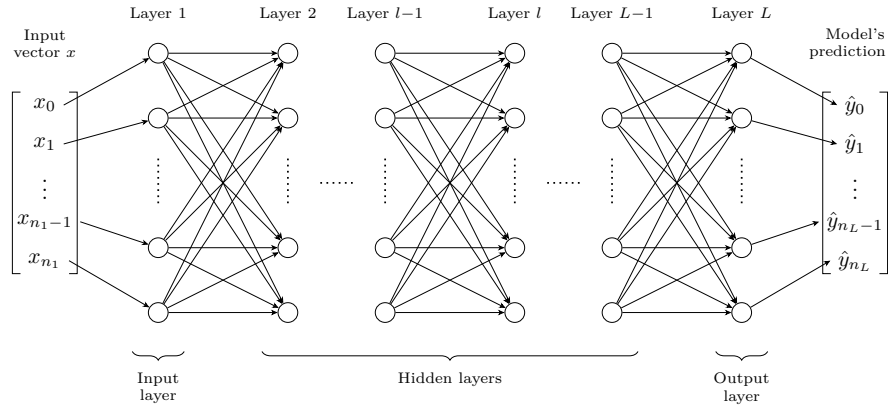


Figure 1.3: Multilayer Perceptron Architecture.

- w_i is the weight associated with the i -th incoming connection,
- b is the bias,
- σ is the activation function.

Figure 1.2 shows a schematic view of this basic unit. The set of weights $w_1, \dots, w_i$ with the bias b constitute the parameters of the neuron, and they are the elements that evolve during training. The magnitude of a weight w (relative to the others) indicates how strongly a neuron depends on the activation of the neurons to which it is connected. A large positive weight means that a high activation in the connected neuron tends to increase the activation of the receiving neuron, whereas a negative weight implies the opposite effect. In this way, the weights modulate the strength and nature of the connections between neurons, thereby defining the overall pattern of interaction within the network.

The activation function σ models how a neuron transforms incoming signals into its output activation. Originally, this function was inspired by biological neurons, with the goal of mimicking the firing behavior of organic cells. However, modern DL no longer relies on this biological interpretation. In practice, any function that is nonlinear, differentiable, and computationally efficient can be used. Nonlinearity is a crucial property, since without it a NN would reduce to a composition of linear transformations, and would therefore be unable to approximate complex functions.

1.3.2 Layers

In MLP architectures, neurons are organized into successive layers, as illustrated in Figure 1.3. This layered structure determines how neurons are connected. In this setting, each neuron receives inputs from all neurons in the preceding layer and sends its output to all neurons in the following layer. Consequently, neurons within the same layer do not interact with each other.

Three types of layers are distinguished:

- an input layer, which directly receives input,

- one or more hidden layers (also referred to as intermediate layers), which perform intermediate transformations,
- an output layer, which produces the final prediction.

The input and output layers play a specific role. The input layer has no preceding layer: its neurons do not compute activations but instead directly encode the components of the input vector x . Conversely, the output layer has no subsequent layer, and its neurons produce the model's prediction $\hat{y}$. This structure imposes a direct correspondence between the dimensionality of the input space and the number of neurons in the input layer, and similarly between the output space and the number of neurons in the output layer.

When describing NN, especially in MLP, it is more common to reason at the level of layers rather than individual neurons. This perspective allows interactions between layers to be modeled in a matrix form. Such a representation enables efficient parallel computation and forms the basis of modern DL implementations [Krizhevsky et al., 2012]. Within this framework, it is useful to introduce the notion of layer activations. The activation of a layer corresponds to the activations of all its neurons, organized into a vector. For a layer l composed of n_l neurons, the activations are given by:

$$\mathbf{h}^{(l)} = [a_1^{(l)}, \dots, a_{n_l}^{(l)}].$$

Using this formulation, the interaction between two consecutive layers can be written compactly as:

$$\mathbf{h}^{(l)} = \sigma^{(l)} \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right),$$

where:

- $\mathbf{h}^{(l-1)}$ represents the activations vector of the previous layer,
- $\mathbf{W}^{(l)} \in \mathbb{R}^{n_l \times n_{l-1}}$ is the weight matrix connecting layer $l-1$ to layer l ,
- $\mathbf{b}^{(l)} \in \mathbb{R}^{n_l}$ is the bias vector associated with layer l ,
- $\sigma^{(l)}$ is the activation function applied element-wise to the neurons of layer l .

This matrix-vector representation is visualised in Figure 1.4, which highlights how each neuron in layer l receives weighted signals from all neurons in the previous layer.

It is important to examine the role played by the intermediate layers of a NN. At least one hidden layer is required to represent sufficiently complex functions. In practice, however, modern architectures often include many more. Theoretical results have shown that, for a fixed number of parameters, increasing the number of layers enhances the expressive capacity of NNs [Telgarsky, 2016]. Each layer is seen as a processing step, and more complex target functions typically require more such steps to be well approximated. However, this interpretation is based on experimental findings rather than a theoretical justification. An entire chapter of this manuscript is devoted to studying the information that can be extracted from the activations of these intermediate layers (Chapter 2).

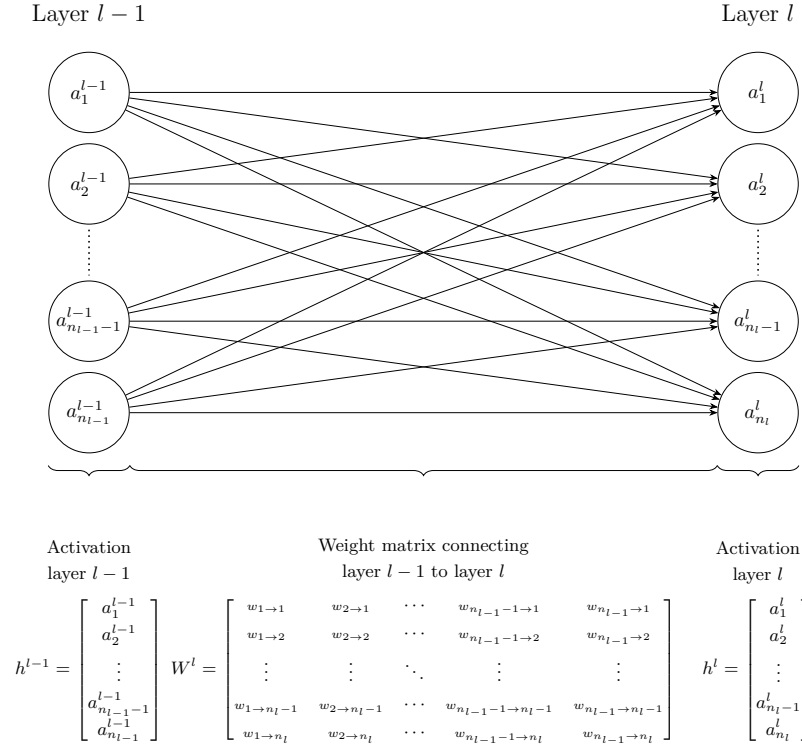


Figure 1.4: Interaction between layers.

1.3.3 Network

With a layer-wise representation, the propagation of information can be described as a sequence of transformations applied from the input layer (indexed 1) to the output layer (indexed L):

$$\hat{f}(x) = \mathbf{h}^{(L)} = \sigma^{(L)} \left(\mathbf{W}^{(L)} \sigma^{(L-1)} \left(\dots \sigma^{(1)} \left(\mathbf{W}^{(1)} x + \mathbf{b}^{(1)} \right) \dots \right) + \mathbf{b}^{(L)} \right) = \hat{y}.$$

For a more compact description of information propagation, the layer activations are defined recursively as:

$$\mathbf{h}^{(l)} = \begin{cases} x & \text{if } l = 1, \\ \sigma^{(l)} \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right) & \text{if } 2 \leq l \leq L. \end{cases}$$

To simplify notation, it is convenient to group all parameters of the network into a single set. The model parameters are thus denoted by:

$$\theta = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^L.$$

In this manuscript, θ is considered as a vector in $\mathbb{R}^P$, where P denotes the total number of parameters, such that $\theta = [w_1, \dots, w_P]$. The notation $\hat{f}_\theta$ expresses the dependence of the NN on its parameters. Increasing the number of parameters, either by adding more neurons per layer or by increasing the

number of layers, can enhance the expressive power of the model, allowing it to approximate more complex functions. However, this also increases computational cost and training time.

It is important to distinguish between the architecture and the parameters of a NN. When referring to the architecture of $\hat{f}$, the number of layers, the number of neurons per layer, and the choice of activation functions are considered. The parameters correspond to θ , which controls the strength of the connections between neurons.

1.4 Learning

Now that the structure of a NN has been detailed, attention must be turned to how it is modified in order to accomplish the assigned task. To do so, attention must be focused on what lies at the very core of DL: the learning process itself. As a reminder, the objective of a NN is to adjust its parameters so that it becomes a good approximation of a target function. That is, for a given architecture $\hat{f}$, the goal is to find the set of parameters θ that best approximates the target function f . Mathematically, this objective is expressed as:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\hat{f}_{\theta}(x^{(i)}), y^{(i)}).$$

However, the NN $\hat{f}$ used to approximate the target function f is highly complex. It involves a large number of parameters as well as many nonlinear components. As a result, an exact solution cannot be found by directly studying the properties of the function. So, one must therefore rely on approximate methods to optimize NNs.

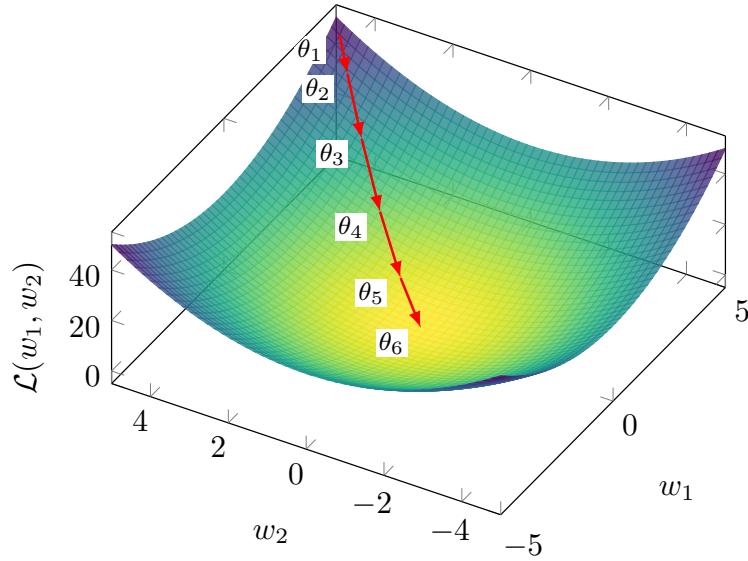
The class of algorithms used for this optimization is known as gradient descent methods. To develop an intuition, Figure 1.5 illustrates gradient descent applied to the simple convex loss function $\mathcal{L}(w_1, w_2) = w_1^2 + w_2^2$, where the red path shows the parameter updates converging to the minimum.

For simplicity, the focus here is on Stochastic Gradient Descent (SGD) with a batch size of 1. As in previous cases, this choice is made because this setting captures the essential intuition behind the training method.

The first step of SGD is to randomly initialize the parameter vector $\theta \in \mathbb{R}^P$ for a given network architecture $\hat{f}$. We denote this initial state by $\theta^{(1)}$. The goal is then to iteratively modify this vector in order to improve the network's ability to approximate the target function.

To determine how to update the parameters, the effect of small variations in each parameter on the loss $\mathcal{L}$ is studied. To analyze how parameter variations affect the loss, a fundamental mathematical tool known as the gradient is used. It is defined as the collection of partial derivatives of a function with respect to its variables. The gradient measures the effect of a small variation of each parameter on the loss.

In our setting, the objective is to understand, for a given example $(x^{(i)}, y^{(i)})$, how the parameters should be adjusted so that the NN prediction $\hat{y}^{(i)}$ becomes closer to the true output $y^{(i)}$. To achieve this, the partial derivatives of the loss

Figure 1.5: Gradient descent in a case $\mathcal{L}(w_1, w_2) = w_1^2 + w_2^2$.

function $\mathcal{L}$ with respect to each parameter vector θ are computed:

$$\nabla_{\theta} \mathcal{L}(\hat{f}_{\theta}(x^{(i)}), y^{(i)}) = \left[\frac{\partial \mathcal{L}(\hat{f}_{\theta}(x^{(i)}), y^{(i)})}{\partial w_1}, \dots, \frac{\partial \mathcal{L}(\hat{f}_{\theta}(x^{(i)}), y^{(i)})}{\partial w_P} \right].$$

Since both the NN and the loss function are compositions of differentiable functions, these derivatives can be efficiently computed using the chain rule. Each component of the gradient $\frac{\partial \mathcal{L}(\dots)}{\partial w_k} \forall k \in [1, \dots, P]$ indicates how a small increase in the corresponding parameter w_k affects the loss. More precisely, for a given parameter, a positive value in the corresponding gradient component indicates that increasing this parameter will locally increase the loss. Conversely, a negative value indicates that increasing it will locally decrease the loss.

Now that information is available on how changes in the parameters affect the loss, the objective is to minimize it. To do so, the parameters are updated in the direction opposite to the gradient. Indeed, the gradient is followed in the opposite direction because it indicates the direction in which the loss increases, whereas the objective is to reduce it. This leads us to the gradient descent update rule:

$$\theta^{(n+1)} = \theta^{(n)} - \eta \nabla_{\theta^{(n)}} \mathcal{L}(\hat{f}_{\theta^{(n)}}(x^{(i)}), y^{(i)}),$$

where $\eta > 0$ is the learning rate, which controls the step size. This parameter must remain small, as the gradient only provides reliable information in a local neighborhood of the current parameters. As a result, a single step is not sufficient to significantly reduce the loss, so the process is iterated many times. Producing a sequence of parameter vectors $\theta_1, \theta_2, \dots$ that progressively improve the model. In practice, training is stopped when successive updates no longer produce a significant decrease in the loss.

It has been mathematically shown that gradient descent applied to a convex loss with respect to the parameters converges to a global optimum, independently

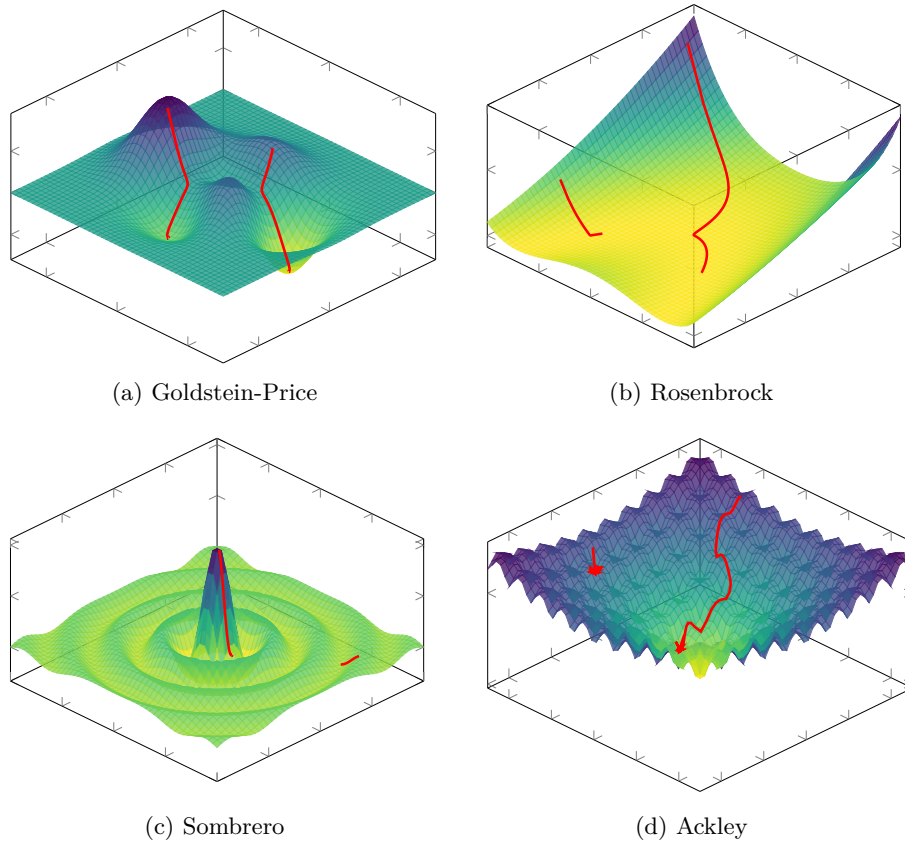


Figure 1.6: Comparison of loss surfaces and gradient descent trajectories.

of the initialization. However, NN training involves highly non-convex loss surfaces, so the optimization trajectory becomes strongly dependent on the initial conditions, as illustrated in Figure 1.6, where different starting points lead to distinct local minima on standard non-convex functions.

Despite the lack of theoretical guarantees of convergence to a global optimum in this setting, the simple procedure of random initialization followed by iterative gradient-based updates is sufficient in practice to train highly complex NNs. This empirical success is often attributed to the expressive power of NNs, even though a complete theoretical understanding of this phenomenon remains an open question.

1.5 Evaluation

Now that the process of training a model has been described, the question of how to evaluate it must be addressed. In the supervised learning setting, this evaluation relies on two aspects:

- its performance on the data it was trained on,
- its performance on data it was not trained on.

The first aspect measures how well the loss minimization process has succeeded. The second measures how well the model generalizes. Indeed, the dataset $\mathcal{D}$ represents a small subset of all possible inputs. To estimate how the model will perform in general, it is important to evaluate how it performs on data it has not seen during training.

In order to quantify these two aspects, the dataset is split into two subsets:

- $\mathcal{D}_{\text{train}}$ is the set on which the model minimizes its loss during the training.
- $\mathcal{D}_{\text{test}}$ is kept aside during training in order to evaluate the model's ability to generalize.

When training is completed, the first step is to verify that the model performs well on the data it was trained on. To do so, the average loss on the training set, denoted $\bar{\ell}_{\text{train}}$, is studied:

$$\bar{\ell}_{\text{train}} = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(x^{(i)}, y^{(i)}) \in \mathcal{D}_{\text{train}}} \mathcal{L}(\hat{f}_{\theta}(x^{(i)}), y^{(i)})$$

If $\bar{\ell}_{\text{train}}$ is not sufficiently low, this means that the model has not been able to learn the task. It is not necessary to go further in the analysis of the model's quality. Before restarting the training process, various design choices are then adjusted. Based on what has been discussed so far in this chapter, these include modifications to the number of layers, the number of neurons per layer, activation functions, the learning rate, and the initialization strategy. However, in practice, there are many other possible variations. There is no universal rule for choosing the appropriate value for each of them. They are therefore chosen empirically, based on models that perform well on similar tasks, combined with intuition. Their selection is often costly and remains a challenging problem in DL.

Assuming now that the model achieves satisfactory performance on the training dataset $\mathcal{D}_{\text{train}}$. We must now study its ability to generalize. To do so, the corresponding average loss on $\mathcal{D}_{\text{test}}$ is computed:

$$\bar{\ell}_{\text{test}} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(x^{(i)}, y^{(i)}) \in \mathcal{D}_{\text{test}}} \mathcal{L}(\hat{f}_{\theta}(x^{(i)}), y^{(i)})$$

If $\bar{\ell}_{\text{test}} \gg \bar{\ell}_{\text{train}}$, the model fails to generalize. This situation most often arises due to a phenomenon known as overfitting. In such cases, the model does not learn general patterns but instead memorizes the training examples. This typically occurs when the number of parameters is too large relative to the amount of training data. The simplest remedy is therefore to reduce the number of parameters in the model. However, it is still necessary to ensure that the model remains sufficiently expressive to perform well on the data.

If $\bar{\ell}_{\text{test}} \approx \bar{\ell}_{\text{train}}$, this indicates that the model generalizes well beyond the training data [Kawaguchi et al., 2022]. The model can therefore be used with confidence for the task on which it was trained.

The ability of a neural network to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations emerge progressively through the hidden layers during training. Understanding their structure is therefore a key

365 step toward understanding how neural networks generalize. The collection of
366 these representations is referred as the Latent Space (LS). The next chapter is
367 devoted to the study of these LSs.

Chapter 2

Latent Space Analysis

The ability of a Neural Network (NN) to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations play a central role in the network's ability to solve a task. Understanding their structure is therefore a key step toward understanding how NN achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep Learning (DL) introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, NN gradually construct representations that become increasingly relevant to the task under consideration [Rumelhart et al., 1986]. These representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training. Since these representations are learned internally by the network and are not directly observable from the input or output spaces, they are commonly referred to as Latent Representation (LR). The collection of these LR learned throughout the network is commonly referred to as a Latent Space (LS).

These LS are now at the core of a growing number of practical applications. Neural video compression exploits the compactness of LS to encode information efficiently [Lu et al., 2019]. In Reinforcement Learning (RL), agent internal representations are used for efficient exploration in extremely open environments [Burda et al., 2018]. Recommendation systems model users and content within a shared LS to compute meaningful similarities [He et al., 2017]. Retrieval-augmented generation systems index and query knowledge bases through latent embeddings [Lewis et al., 2021]. Finally, Large Language Model (LLM) and multimodal architectures rely on the quality of learned representations to align heterogeneous modalities such as text, images, and audio [Radford et al., 2021]. These examples represent only some of the most widespread applications of LR.

Given their extensive use, the study of LR has become a central topic in DL research. A major objective is to understand how information is organized within these spaces, which concepts are encoded, and how these representations

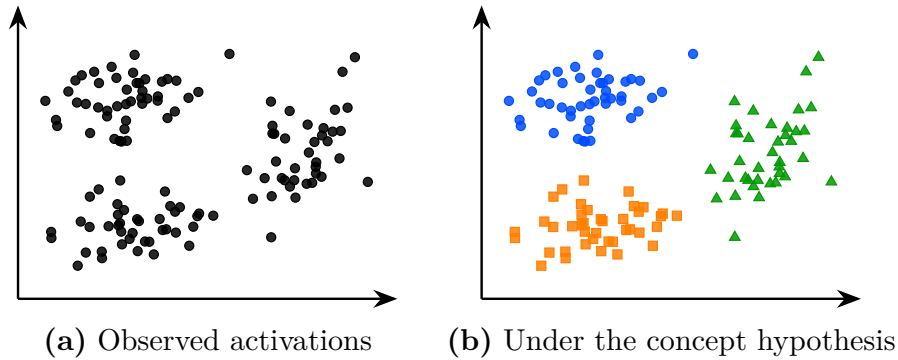


Figure 2.1: The same latent space activations seen without concept hypothesis (a) and under the working hypothesis that concepts emerge in the latent space (b). Colors represent hypothetical concept memberships assigned by the model; they are not observed labels. Under this hypothesis the structure of the latent space becomes meaningful and amenable to analysis.

influence model behavior.

In this chapter, we first make explicit the working hypothesis that underlies much of the research on LS analysis. Throughout this manuscript, we refer to this assumption as the concept hypothesis 2.1. Once this conceptual framework has been established, we review the main approaches proposed in the literature to identify and extract concepts from LR. To the best of our knowledge, the main families of methods include probing methods (Section 2.2), semantic directions (Section 2.3), sparse autoencoders (Section 2.4), and clustering-based approaches (Section 2.5). Finally, Section 2.6 draws a comparative synthesis of these methods and discusses the open questions that remain.

2.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on LS analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, LS, concept, and semantics. The goal of this section is to make this conceptual framework explicit by defining its main components.

This hypothesis originates from the need to explain the generalization ability of deep NN observed after training, as discussed in Section 1.5. A model is said to generalize when, for an input $x \in \mathcal{X}$ not seen during training, it still produces an output $\hat{y}$ close to the true output y . Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 2.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task [Bengio et al., 2014].

Definition 2.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction or decision-making.

The ability of NN to perform abstraction arises from their intermediate activations. Indeed, a NN does not directly map x to y . Instead, as illustrated in Figure 1.3, it computes a sequence of intermediate activations across hidden layers. These activations are numerical encodings of the input referred to as embeddings. Once training is complete, these embeddings have been structured so as to be capable of performing a specific task. To mark this distinction, we will refer to these trained embeddings as latent representations, as defined in Definition 2.1.2.

Definition 2.1.2: Latent Representation

A representation is the internal encoding of an input given by a intermediate layer of a trained NN.

Each hidden layer refines the representation of the previous layer. Each transformation progressively reshapes the representation into a form more directly aligned with the output space [Rumelhart et al., 1986]. This process can be described as a sequence of transformations where each $h^{(l)}$ is a representation:

$$x \rightarrow h^{(1)} \rightarrow h^{(2)} \rightarrow \dots \rightarrow h^{(L-1)} \rightarrow \hat{y}$$

The set of all such representations produced by a layer over a dataset defines its LS, as defined in Definition 2.1.3.

Definition 2.1.3: Latent Space

The LS is the set of all representations produced by a layer over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

Under the concept hypothesis, the analysis of the LS is assumed to provide access to higher-level abstract structures learned by the model, as illustrated in Figure 2.1. These structures are referred to as concepts. In this manuscript, a concept is defined as an abstract representation that organizes information and supports reasoning (see Definition 2.1.4). Its relevance is therefore determined not by whether it is “true,” but by whether it provides a useful description for the task under consideration.

Our definition of concepts is consistent with both the pragmatic tradition in philosophy and the information-processing view of cognition. Within these frameworks, concepts arise from the need to simplify an otherwise intractable reality. Because the environment contains more information than any cognitive system can process, intelligent behavior relies on abstractions, called concepts, that discard irrelevant details while preserving the information required to achieve a given objective [Peirce, 1878, Misak, 2013]. These concepts are realized as internal representations that reduce complexity and enable reasoning under limited computational resources [Newell, 1980, Simon, 1996]. Although these

theories were originally developed to explain human cognition, the concept hypothesis assumes that the same principles can be extended to deep NN.

In order to influence the model’s computations, concepts must necessarily admit a concrete realization within the LS. We refer to this realization as the concept structure, or simply the structure. The definition of the structure is provided in Definition 2.1.5. The methods reviewed in this chapter all subscribe to the concept hypothesis, but they disagree on what structure a concept should take. This choice necessarily affects the extraction procedure, which is why several families of LS analysis methods coexist in the literature.

Definition 2.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

Definition 2.1.5: Concept Structure

A concept structure is the specific form for how a concept is instantiated within the LS.

Now that the definition of a concept has been established, we can consider its relationship to human. According to our definition, we can imagine concepts that are useful for the machine but do not make sense to humans. This relationship is captured by the notion of semantics [Marconato et al., 2023], defined in Definition 2.1.6. Some concepts developed in LS can be readily associated with human concepts and are said to have high semantic value. Others may contribute to task performance while remaining difficult for humans to understand, and are said to have low semantic value.

Definition 2.1.6: Semantics

Semantics refers to the alignment between concepts in the model LS and concepts used by a human observer. A concept is semantic if it can be consistently associated with a human-interpretable concept.

It is important to note that this conceptual framework is not supported by a complete theoretical understanding of deep NN. Current mathematical tools do not provide a rigorous characterization of their internal representations and do not formally prove that concepts emerge as identifiable structures within LS. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of representation and LS analysis.

The remainder of this chapter is devoted to reviewing methods for analyzing LS and extracting the concepts they encode. We begin with probing, one of the most direct approaches for assessing whether concepts are represented in the LS.

2.2 Probing

Probing methods aim to determine whether a given semantic concept is encoded within the LS of a NN [Li et al., 2024, Karvonen, 2024]. More specifically, they investigate whether the presence or absence of a selected concept can be inferred

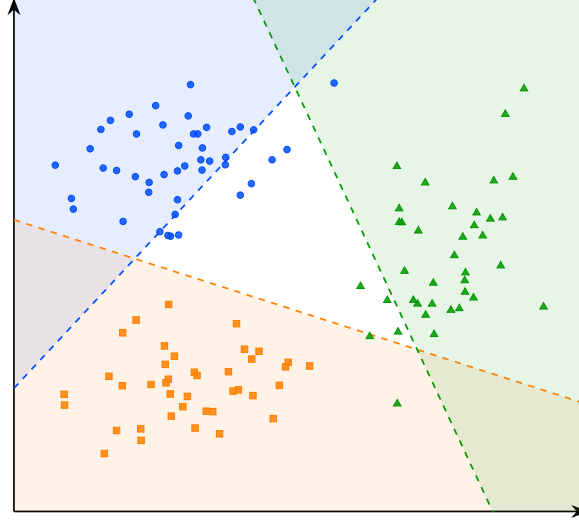


Figure 2.2: Decision boundaries of the linear probes for concepts A, B, and C. The colored regions indicate the areas classified positively by each probe.

from embeddings. The underlying assumption is that if a concept can be reliably decoded from these representations, then the network likely encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of semantic concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by c_j the presence or absence of concept j in example x . The dataset can therefore be written as follows:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}, c_1^{(i)}, \dots, c_n^{(i)})\}_{i=1}^N.$$

The data are subsequently projected into the LS of the pretrained model. We denote by h^l the representation vector obtained at layer l . From these LR, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe and is denoted by p . Its purpose is to determine whether a given concept is present in the representation. Its purpose is therefore to assess whether information associated with the concept is accessible from the network's internal representations. This can be formalized as follows:

$$p_{\theta}^{l,j} : h^l \mapsto c_j, \quad \theta^* = \arg \min_{\theta} \mathcal{L}(p_{\theta}^{l,j}(h^l), c_j)$$

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to information explicitly encoded in the LS. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the NN and

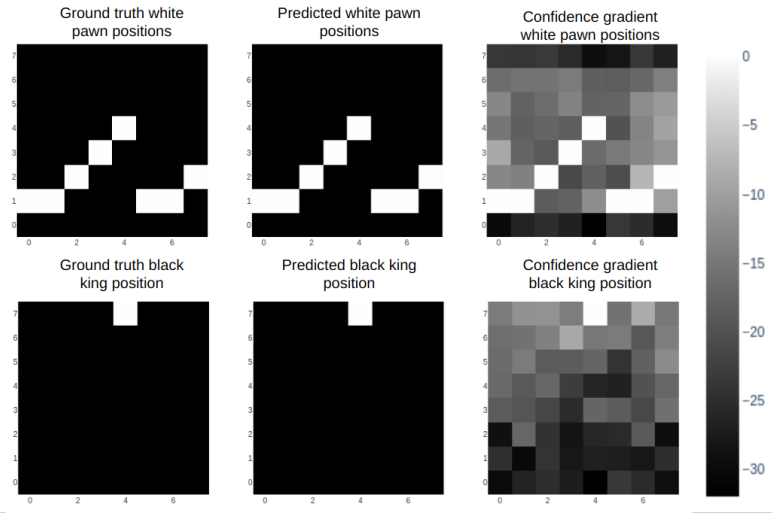


Figure 2.3: Probing analysis of a large language model trained to play chess [Karvonen, 2024]. Each row corresponds to a piece type: white pawns (top) and the black king (bottom). The left column shows the ground truth board positions, the middle column shows the positions predicted by the probe from the model’s internal representations, and the right column shows the confidence gradient across board squares. The close agreement between ground truth and predicted positions suggests that the model internally maintains a structured representation of the board state, even though it was only trained on sequences of moves expressed in natural language.

potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the LS contains exploitable information related to the considered concept.

An application of this technique can be found in studies investigating whether LLM develop internal representations of their environment [Li et al., 2024, Karvonen, 2024]. In these works, researchers train LLM to play games such as chess or Othello in a purely textual setting. The model is provided only with sequences of moves and information about the actions it can take, all expressed in textual form. The central question is whether the LLM develops an internal representation of the game state. To address this question, the probed concepts correspond to the presence or absence of different game pieces on specific board positions. As illustrated in Figure 2.3, probes are able to reconstruct the full board state from the model’s hidden representations, even though the only available information to the model is a sequence of text.

However, the fact that a concept can be decoded from LR does not necessarily imply that the model actually uses this information to perform its task. To assess whether representations play a causal role in the model’s output, it is possible to perform interventions on the LR. An intervention consists of modifying a LR in a controlled manner by activating or deactivating a given concept, and then studying its impact on the model’s output. If the resulting change in the model’s output is consistent with the applied perturbation, this suggests that the representation had a causal role.

To this end, the LR h^l is minimally modified such that the probe prediction for the target concept c_j changes state, while the predictions associated with other concepts remain unchanged. This goal can be formulated as an optimization problem over the latent activation itself. The optimization is performed directly on the representation $h^{l'}$, using gradient-based optimization as introduced in Section 1.4. The solution to this problem is the optimal modified representation h^{l*} , defined as:

$$h^{l*} = \arg \min_{h^{l'}} \underbrace{\gamma \mathcal{L}_{\text{flip}}(p_{\theta}^{l,j}(h^{l'}), c_j)}_{\text{flip target concept } c_j} + \underbrace{\lambda \sum_{k \neq j} \mathcal{L}_{\text{preserve}}(p_{\theta}^{l,k}(h^{l'}), c_k)}_{\text{preserve other concepts } c_{k \neq j}} + \underbrace{\beta \|h^{l'} - h^l\|_2^2}_{\text{minimality}}$$

where:

- h^l is the original LR of the input at layer l ,
- $h^{l'}$ is the modified LR being optimized,
- h^{l*} is the optimal modified representation,
- $\mathcal{L}_{\text{flip}}$: loss encouraging the prediction of the target concept c_j to change state (i.e., flipping the predicted label),
- $\mathcal{L}_{\text{preserve}}$: loss enforcing invariance of all non-target concepts c_k for $k \neq j$,
- $\|h^{l'} - h^l\|_2^2$: regularization term ensuring the modified representation remains close to the original LR,
- γ, λ, β : weighting coefficients controlling the trade-off between the intervention strength, the preservation of other concepts, and the minimality of the perturbation.

In the context of LLM studied previously, such interventions have been used to remove specific pieces from the model’s internal representation of the game state [Li et al., 2024, Karvonen, 2024]. The main observation is that, in the vast majority of cases, the model behaves as if the piece had effectively disappeared. The LLM continues to generate legal moves while correctly accounting for the absence of the removed piece. This provides evidence for the causal role of these internal representations in the model’s behavior.

A main limitation of the probing approach is that the concepts under investigation must be predefined before any analysis can be performed. This introduces a strong bias in what can be studied. Moreover, annotating a dataset for each concept is extremely time-consuming and requires significant human effort. Interventions in LS are also difficult to interpret, as it is not always clear what constitutes a meaningful change in the network’s output under a given perturbation.

Moreover, probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a LR. However, this binary perspective can be overly restrictive, as concepts may be represented in a more continuous or graded manner, with varying degrees of presence. To obtain a more detailed characterization of these representations, other approaches have been proposed, such as the analysis of interpretable directions in LS.

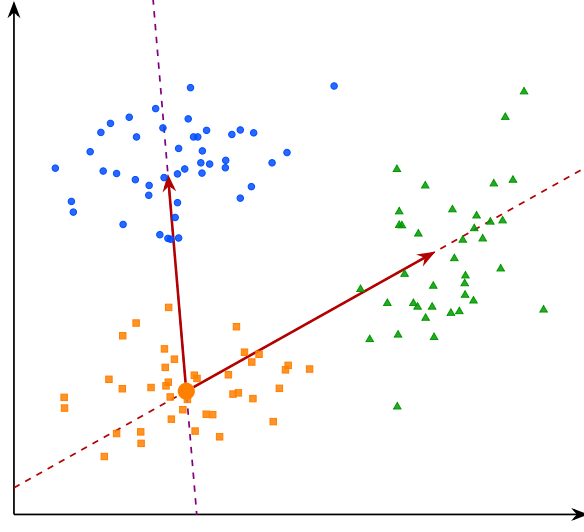


Figure 2.4: Interpretable directions in the latent space. The arrows represent semantic shifts from concept B toward concepts A and C . Moving along these directions in the latent space induces a coherent change in the model’s behavior, suggesting that the transition between concepts is continuously encoded along these axes.

2.3 Interpretable Directions

To overcome the limitations of binary concept detection in probing methods, several approaches have been proposed to analyze the structure of LR. Among them, the notion of interpretable directions has emerged as a natural extension, aiming to capture the continuous nature of how concepts are encoded in representation spaces. Empirically, it has been observed that shifting a LR along a direction can induce coherent and semantically meaningful changes in the model’s output [Haas et al., 2024, Voynov and Babenko, 2020]. Such transformations can be mathematically express as:

$$h' = h + \alpha \mathbf{v}_i,$$

where:

- h denotes the original embedding,
- $\mathbf{v}_i$ is a unit vector associated with concept i ,
- α controls the strength of the intervention,
- h' is the resulting perturbed representation.

In many cases, the resulting change in the model’s output varies smoothly with α , suggesting an approximately linear structure of the LS with respect to certain semantic factors, as illustrated in Figure 2.5. This observation has led to the hypothesis that neural representations may organize concepts in a partially linear manner within the embedding space. Vectors of this form are referred to as interpretable directions, as they correspond to transformations that can be

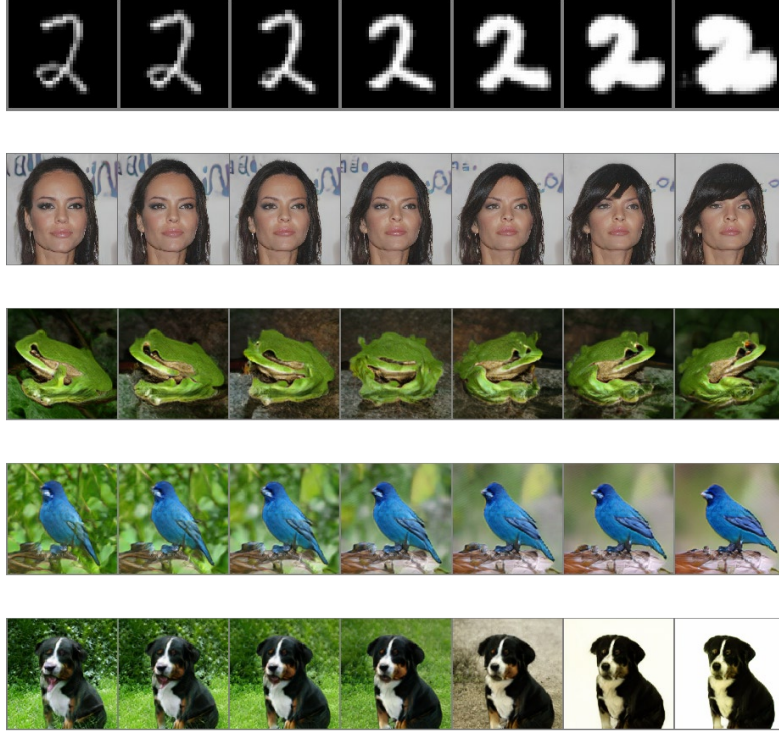


Figure 2.5: Interpretable directions in the latent space of a generative model [Voynov and Babenko, 2020]. Each row corresponds to a distinct direction $\mathbf{v}_i$, and each column to an increasing value of α . The resulting transformations are semantically coherent and diverse: stroke thickness for digits, head rotation for faces, texture and background for natural images.

601 associated with specific semantic attributes. The objective of this line of work is
 602 therefore to identify directions in LS that align with meaningful variations in
 603 the encoded concepts.

604 This phenomenon has been extensively studied in generative image models.
 605 As illustrated in Figure 2.5, controlled perturbations of LR along interpretable
 606 directions can induce specific semantic transformations, such as stroke thickness
 607 modification in digit images, head rotation in facial images, or texture and
 608 background changes in natural images [Voynov and Babenko, 2020]. These ob-
 609 servations have led to the development of the field of LS editing, whose objective
 610 is to manipulate generated content directly through modifications in LS rather
 611 than through changes in the input prompt.

612 However, a central challenge remains: identifying such interpretable directions
 613 is far from trivial. Most existing approaches rely on auxiliary models that encode
 614 human-defined semantics, such as CLIP [Haas et al., 2024], to relate latent
 615 variations to textual concepts. Consequently, these methods inherit a limitation
 616 similar to that of probing techniques 2.2, since concept discovery depends on a
 617 pre-existing semantic framework defined by humans.

618 To overcome this limitation, a research direction known as unsupervised

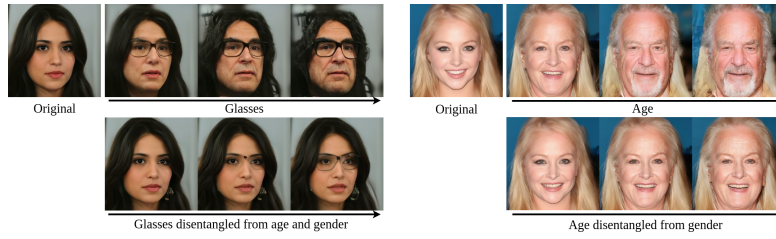


Figure 2.6: Illustration of entanglement in the latent space of a generative face model [Haas et al., 2024]. The top rows show entangled directions: moving along the glasses direction simultaneously modifies age and gender, and moving along the age direction also alters gender. This reflects the entanglement phenomenon, where a single direction in latent space induces unintended changes across multiple semantic concepts at once. The bottom rows show the corresponding disentangled directions, where each axis controls a single semantic attribute while leaving the others unchanged.

semantic discovery has emerged. Its goal is to uncover interpretable directions in LS without explicit semantic supervision. In these approaches, one model applies perturbations along random directions in LS, while a second model attempts to recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent structure in LS [Voynov and Babenko, 2020]. In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations.

However, despite these advances, several challenges remain. It appears that not all concepts encoded by neural networks are linearly identifiable in LS. In practice, interpretable direction methods typically discover only a few dozen directions per dataset, which intuitively seems insufficient to account for the full diversity of semantic variations that a generative model is capable of producing.

In addition, LR often exhibit a phenomenon known as entanglement, illustrated in Figure 2.6, where multiple semantic attributes are mixed within the same directions of the representation space. Ideally, one would like each semantic factor to vary independently along a specific direction, without affecting other attributes. In practice, this is rarely the case: a single direction in LS often influences multiple semantic simultaneously, meaning that modifying one concept unintentionally alters several others. This lack of separability indicates that semantic information is not cleanly factored in the representation space, which motivates the search for more structured decompositions.

The limited number of linearly identifiable concepts, combined with the difficulty of isolating independent semantic factors, has motivated a stronger hypothesis about how neural networks organize information internally: the superposition hypothesis. In this context, a preferred analytical tool for investigating this hypothesis is the use of Sparse AutoEncoders (SAE), which can extract concepts from neural representations.

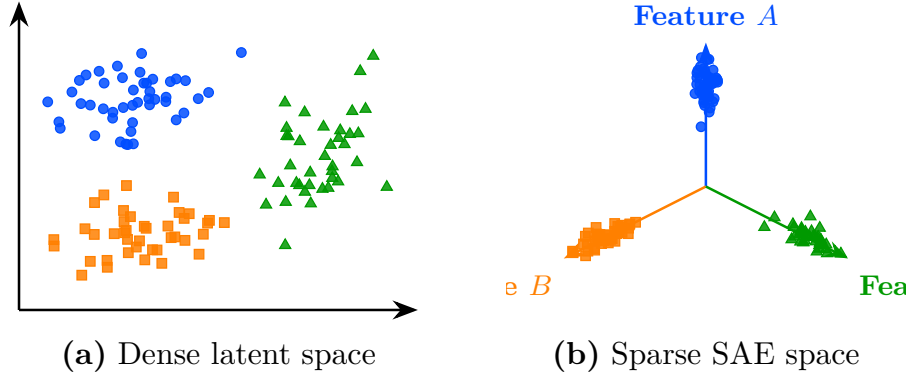


Figure 2.7: Transformation from a dense latent space to a sparse SAE-style representation. On the left (a), embeddings are entangled in the initial latent space. On the right (b), the sparse representation aligns each concept with a dedicated feature axis.

2.4 Sparse Autoencoder

Approaches based on latent directions face inherent limitations. Empirically, only a restricted subset of concepts appears to admit a linear representation, and even these directions are often affected by significant entanglement with other concepts, illustrated in Figure 2.6.

To explain this phenomenon, a complementary hypothesis has been introduced, extending the assumptions underlying the Concept Hypothesis (2.1). This hypothesis, known as the Superposition Hypothesis [Elhage et al., 2022], posits that concepts are preferentially represented in a linear fashion, with each concept ideally corresponding to a distinct direction in LS. In this idealized setting, concept vectors would form an approximately orthogonal basis, where each direction independently controls a single factor of variation. Furthermore, only a small subset of the available concepts is expected to be active for a given representation. As a result, LR can be viewed as sparse combinations of concept directions. This idealized representation is referred to as a non-superposed representation and can be mathematically expressed as follows:

$$\mathbf{h} = \sum_{i=1}^n \alpha_i \mathbf{v}_i, \quad \forall i \neq j, \quad \mathbf{v}_i^\top \mathbf{v}_j \approx 0, \quad \|\boldsymbol{\alpha}\|_0 \ll n,$$

where:

- $\mathbf{h}$ denotes a LR,
- n is the total number of concepts in the dictionary,
- $\mathbf{v}_i$ represents the direction associated with concept c_i ,
- α_i corresponds to the activation strength of concept c_i in the representation $\mathbf{h}$,
- $\mathbf{v}_i^\top \mathbf{v}_j \approx 0$ indicates that concept directions are approximately orthogonal, meaning each concept independently controls a single factor of variation,



Figure 2.8: A concept direction corresponding to the Golden Gate Bridge, discovered by a sparse autoencoder applied to a large language model [Templeton et al., 2024]. This direction activates strongly whenever the Golden Gate Bridge is evoked in the input, whether through English text, other languages, or even images, demonstrating that the learned concept is not tied to a specific surface form but captures a deeper semantic abstraction shared across modalities.

- $\|\alpha\|_0 \ll n$ indicates that only a small fraction of concepts are active for a given representation.

According to the Superposition Hypothesis, latent representations do not perfectly follow this idealized structure because the representational capacity of NN is limited. For a given layer, the dimensionality of the representation space is finite, which constrains the number of concepts that can be encoded independently. As a result, the network cannot allocate a dedicated direction to every concept, and multiple concepts end up compressed into shared dimensions [Bricken et al., 2023]. This phenomenon is closely related to the entanglement observed in LS analyses: where entanglement is an empirical observation about the mixing of semantic attributes, the Superposition Hypothesis provides a theoretical explanation for why such mixing emerges.

Under this hypothesis, the phenomenon of superposition becomes an unavoidable obstacle for any method that attempts to analyze LR directly in the original activation space. Motivated by this view, and with the goal of recovering the idealized non-superposed, a large body of work has focused on developing methods to desuperpose LR. To this end, a common strategy consists in projecting LR into a higher-dimensional space while imposing additional constraints designed to recover a non-superposed representation. A widely used implementation of this idea is the SAE, which can be defined as follows:

$$z = f_{\text{enc}}(h), \quad \hat{h} = f_{\text{dec}}(z)$$

$$\mathcal{L}_{\text{auto}} = \gamma \|h - \hat{h}\|_2^2 + \lambda \|z\|_1$$

where:

- $f_{\text{enc}}(h)$ is the encoding function that projects the original representation h into a higher-dimensional sparse space,
- $f_{\text{dec}}(z)$ is the decoding function that reconstructs the original representation from the sparse code,

- 697 • $z \in \mathcal{Z}$ is the sparse representation of h , with $\dim(z) \gg \dim(h)$,
- 698 • $\hat{h}$ is the reconstructed representation,
- 699 • $\gamma \|h - \hat{h}\|_2^2$ is the reconstruction loss ensuring that the original representation
700 can be faithfully recovered,
- 701 • $\lambda \|z\|_1$ is the sparsity regularization term that encourages most components
702 of z to be zero,
- 703 • γ, λ are weighting coefficients controlling the trade-off between reconstruc-
704 tion fidelity and sparsity.

705 In practice, both f_{enc} and f_{dec} are implemented as linear projections into the
706 high-dimensional space $\mathcal{Z}$. This choice is motivated by the same reasoning as
707 in probing methods (Section 2.2). Using overly expressive projection functions
708 would introduce additional bias, making it difficult to attribute the structure
709 of the learned representations to the model itself rather than to the projection.
710 The parameters of these projection functions are learned by minimizing $\mathcal{L}_{\text{auto}}$
711 using gradient-based optimization, as introduced in Section 1.4. Under this
712 formulation, each dimension of $\mathcal{Z}$ is interpreted as a dedicated concept direction.
713 The corresponding basis vector serves as the support for a distinct concept
714 encoded by the model.

715 The method proves to be powerful when successfully applied to real language
716 models [Templeton et al., 2024]. In these applications, the idea is to desuperpose
717 LLM LR using SAE in order to recover a non-superposed representation. Once
718 this representation is obtained, it has been shown that many concept vectors
719 exhibit strong semantic properties. A well-known example is the Golden Gate
720 Bridge concept direction, illustrated in Figure 2.8, which activates systematically
721 whenever the input text or image refers to the Golden Gate Bridge, regardless
722 of the language or modality.

723 It is possible to artificially activate this concept direction by directly modifying
724 the sparse representation z , even when the input does not contain any reference
725 to it. This operation constitutes an explicit intervention on the model’s latent
726 representations. The edited representation is then projected back into the original
727 activation space and the model proceeds with standard next-token generation
728 from h^* :

$$h^* = f_{\text{dec}}(z^*), \quad \text{where} \quad z^* = z + \delta e_i,$$

729 where:

- 730 • z is the original sparse representation of the input,
- 731 • z^* is the modified sparse representation,
- 732 • e_i is the basis vector of $\mathcal{Z}$ associated with the target concept direction,
- 733 • δ controls the strength of the artificial activation,
- 734 • h^* is the resulting edited representation in the original activation space.

735 Empirically, it is observed that even when the prompt is unrelated to the
736 Golden Gate Bridge, the model’s output shifts to discussing it. This provides
737 evidence for the causal role of these learned features. This type of intervention

has been used by Anthropic researchers to steer or suppress specific model behaviors, effectively enabling a form of behavior control through latent-space editing.

Despite their empirical success, SAE exhibit several important limitations that question some of their underlying assumptions. These limitations can be summarized as follows:

- **Instability of the decomposition.** Repeating the SAE training process multiple times on the same model does not yield the same decomposition of the LS. Different runs produce different concept directions in $\mathcal{Z}$, even when achieving comparable reconstruction performance, which raises questions about the uniqueness and reliability of the discovered structure.
- **Lack of semantic coverage.** In the high-dimensional space $\mathcal{Z}$, a large fraction of dimensions do not correspond to any interpretable semantic concept. This abundance of semantically vacuous directions questions the extent to which the decomposition genuinely reflects the conceptual structure of the model.
- **Sparsity–semantics trade-off.** Increasing the sparsity coefficient λ beyond a certain threshold degrades the semantic quality of the learned features [Jermyn et al., 2024]. The representations tend to lose meaningful structure, suggesting that enforcing excessive sparsity comes at the cost of interpretability.
- **Failure of scaling to remove superposition.** Under the Superposition Hypothesis, increasing the dimensionality of the representation space should reduce superposition. However, this is not observed in practice. Even in very high-dimensional LS, representations remain partially superposed, indicating that scaling alone does not resolve the entanglement of features.

These limitations suggest that SAE, while powerful, do not fully resolve the problem of representational structure, and that convincing empirical results do not necessarily validate the underlying theoretical assumptions. Indeed, probing (Section 2.2), interpretable directions (Section 2.3), and SAE all rest on a shared assumption: that semantic structure can be recovered through approximately linear decompositions of LR. This assumption has driven significant empirical progress, yet it remains theoretically fragile. The following section therefore introduces a complementary approach that relaxes this constraint entirely, requiring no assumption about the linear organization of concepts: deep clustering.

2.5 Clustering

The LS analysis methods presented so far all rely, either explicitly or implicitly, on assumptions about the linear organization of concepts within the LS. For probing methods, concepts must be at least partially linearly separable for simple probes to succeed 2.2. Interpretability through latent directions assumes that variations associated with a concept can be captured by moving along a linear axis of the LS 2.3. Similarly, the superposition hypothesis assumes that concepts are represented through linear directions that become entangled because of representational capacity constraints 2.4. Although these assumptions have led

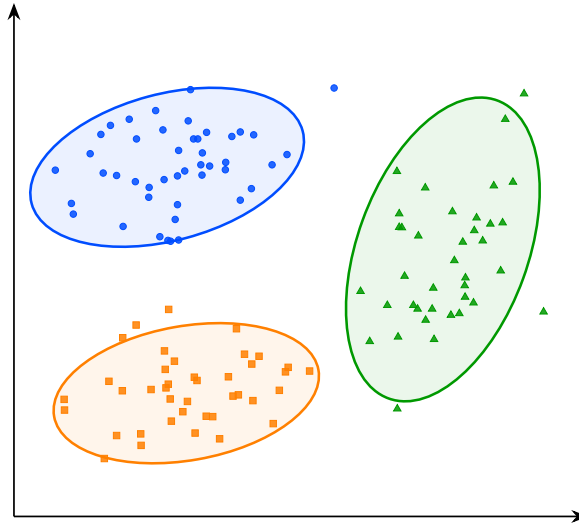


Figure 2.9: Visualization of the latent space structure. The points represent embeddings associated with concepts A, B, and C. The ellipses indicate the approximate extent of each group in the latent space and highlight their geometric separation.

to numerous empirical successes, their theoretical justification remains limited. More importantly, several observations suggest that the linearity hypothesis may not fully capture the structure of LR.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity often degrades reconstruction quality and interpretability, suggesting that the underlying representation may not be perfectly described by a sparse linear decomposition. Taken together, these observations indicate that semantic concepts may not always be organized according to a simple linear structure. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of LS.

To the best of our knowledge, clustering-based analysis is the only family of LS analysis methods that does not rely on any assumption of linear structure, which is why the following section is devoted to its presentation [Ren et al., , Wei et al.,]. Rather than assuming that concepts correspond to directions or linear subspaces, clustering methods only assume that semantically similar inputs produce nearby LR. Conversely, inputs that differ significantly from a semantic perspective are expected to be located farther apart in the LS. This is a considerably weaker assumption, in the sense that it is empirically verified in a wide range of settings.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of embeddings that occupy the same region of the LS. Concepts are therefore no longer associated with directions, but with clusters of neighboring representations, as illustrated in Figure 2.10. This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is first projected into a LS using a NN. The resulting embeddings form a point cloud on which standard clustering algorithms can

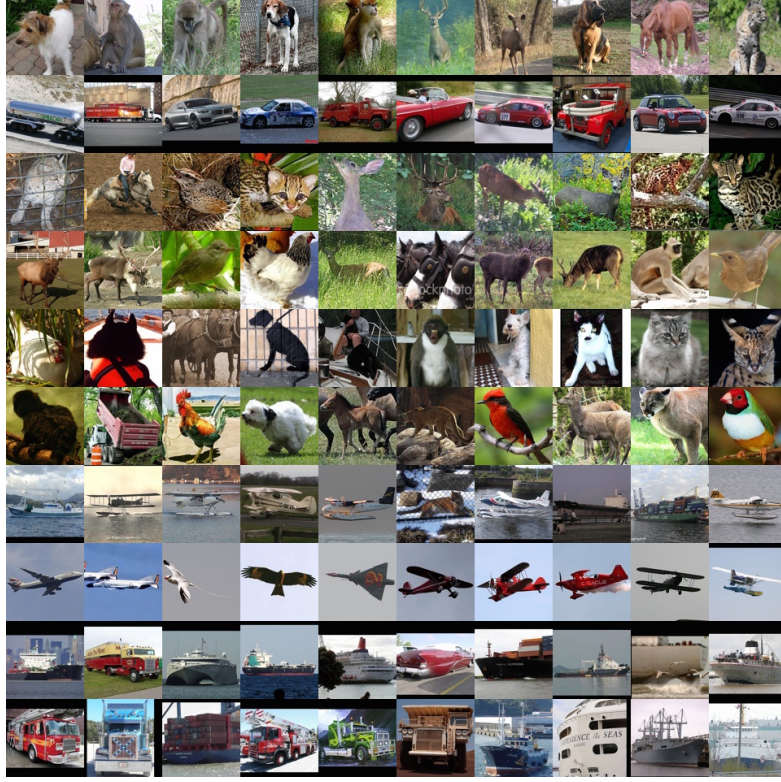


Figure 2.10: Clusters discovered by a deep clustering model applied to natural images [Xie et al.,]. Each row contains the ten highest-scoring elements of a single cluster. The visual coherence within each row, grouping animals, vehicles, and aircraft into distinct clusters, suggests that the learned latent representations capture semantically meaningful structure without any explicit supervision.

809 be applied. The obtained clusters are then interpreted as representing distinct
 810 semantic concepts.

811 Deep clustering has been particularly successful in unsupervised image classi-
 812 fication. Typically, an autoencoder is composed of an encoder f_{enc} and a decoder
 813 f_{dec} . The encoder maps an input $x^{(i)} \in \mathcal{X}$ to a low-dimensional LR $z^{(i)} \in \mathbb{R}^d$:

$$z^{(i)} = f_{\text{enc}}(x^{(i)}), \quad d \ll \dim(\mathcal{X}).$$

814 To ensure that this compressed representation does not discard essential
 815 information, the decoder reconstructs the input from the LS:

$$\hat{x}^{(i)} = f_{\text{dec}}(z^{(i)}) = f_{\text{dec}}(f_{\text{enc}}(x^{(i)})).$$

816 The model is trained to minimize the reconstruction error between the original
 817 input and its reconstruction, typically using a Mean Squared Error (MSE) loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - \hat{x}^{(i)}\|_2^2.$$

This objective encourages the LS to preserve the most informative features of the input while enforcing a compact representation.

Clustering algorithms are then applied to the learned embeddings.

Let $\mathcal{D} = \{x^{(i)}\}_{i=1}^N$ denote a dataset of inputs. Through the encoder f_{enc} , this dataset induces a point cloud in the LS:

$$\mathcal{Z} = \{z^{(i)} \in \mathbb{R}^d \mid z^{(i)} = f_{\text{enc}}(x^{(i)}), x^{(i)} \in \mathcal{D}\}.$$

Clustering algorithms are then applied to this set $\mathcal{Z}$ of embeddings in order to identify groups of semantically similar representations.

An example of this methodology can be observed in Figure 2.10, where the approach is applied to the STL-10 dataset [Coates et al., 2011] using k -means clustering [MacQueen, 1967] with a number of clusters arbitrarily fixed to $K = 10$ [Xie et al.,]. For each cluster, the corresponding line shows the 10 samples closest to the associated centroid in the LS.

We observe that the method captures coherent semantic categories such as animals, vehicles, and flying entities. This result is particularly striking given that the only available information to the system consists of raw images. It suggests that the compression process induces a LS in which semantically meaningful structures naturally emerge, and that these structures correspond to high-level concepts.

More advanced approaches combine representation learning and clustering objectives within a single optimization process. In this setting, additional loss terms encourage embeddings to organize themselves around cluster centroids during training. This often produces LS that are easier to analyze and improves the quality of the discovered semantic structure [Guo et al., , Miklautz et al.,].

Despite its effectiveness, clustering-based analysis presents several important limitations. First, as with most unsupervised semantic discovery techniques, the interpretation of the resulting clusters remains partially subjective: while benchmark datasets often provide class labels that facilitate evaluation, there is no guarantee that the discovered clusters correspond to the categories that humans consider most relevant, and alternative semantic groupings may be equally valid. Second, clustering methods are highly sensitive to design choices, as the number of clusters, the distance metric, and the clustering algorithm itself can all significantly influence the results, making parameter selection challenging and often requiring substantial empirical tuning. Finally, to the best of our knowledge, no intervention technique has yet been proposed in the context of clustering-based analysis: unlike probing or SAE, which allow targeted modifications of LR to study their causal role, clustering methods currently offer no mechanism to alter the model’s behavior by acting directly on the discovered structure.

2.6 Conclusion

The four families of methods reviewed in this chapter, namely probing, interpretable directions, sparse autoencoders, and clustering, are summarized in Figure 2.11 and all build upon the concept hypothesis introduced in Section 2.1. Each of them approaches the question from a different angle, but they share a common objective: to uncover how LR are structured and what semantic information can be extracted from them.

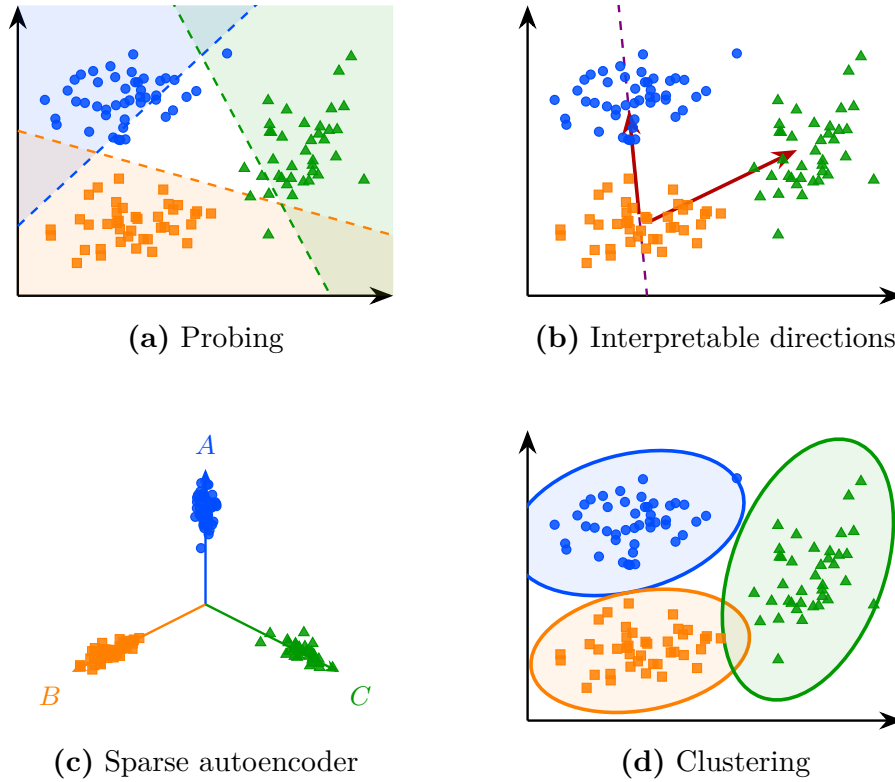


Figure 2.11: Overview of the four latent space analysis methods studied in this chapter, applied to the same point cloud.

However, a fundamental difficulty remains. There is currently no consensus on how to rigorously evaluate to what extent these methods genuinely provide access to the internal concepts of a model. Most existing metrics assess reconstruction quality or linear separability, but these do not directly measure whether the discovered structures correspond to meaningful abstractions from the model's perspective. This leaves open the question of what it means, formally, for a concept to be represented in a NN.

Addressing this question may require contributions from disciplines beyond computer science and mathematics. As the notion of semantics is inherently tied to human interpretation, a purely formal or computational account may be insufficient. Insights from philosophy, cognitive science, or psychology could prove valuable in grounding the notion of concept in a way that is both theoretically rigorous and empirically tractable. Despite these open questions, the methods studied in this chapter provide a practical foundation for analyzing the internal representations of NNs.

Chapter 3

Reinforcement Learning

This chapter introduces the fundamental concepts of Reinforcement Learning (RL), providing the necessary foundation for the ideas developed throughout the rest of this manuscript. Its structure is as follows:

Section 3.1: An overview of the milestones that led to the emergence of RL as a unified field.

Section 3.2: A formalization of the objective underlying RL algorithms.

Section 3.3: An explanation of how policies are progressively refined through interaction with the environment to approach the optimal policy.

3.1 Historical Developments

RL is today presented as a major field within artificial intelligence. However, it originates from several research traditions that progressively converged toward a common framework. In this section, we outline the main stages in the emergence of this field, highlighting the key ideas and milestones that shaped its evolution.

3.1.1 Behavioral foundations (1900–1950)

The origins of RL can be traced back to 20th-century animal psychology studies [Thorndike, 1911]. These studies aimed to understand how behaviors are formed and modified. The central idea emerging from this work is that learning is driven by the consequences of actions. Behaviors followed by positive outcomes are more likely to be repeated, whereas those associated with negative outcomes tend to gradually decrease in frequency.

This perspective was later developed within behaviorism. In this school of psychology, rewards and punishments are used to gradually shape behavior through repeated interaction with the environment [Skinner, 1938]. The term “reinforcement” in the context of learning comes from this school of thought, where it refers precisely to the process of strengthening a behavior through its consequences.

This idea led to the suggestion that learning principles observed in living organisms could be transferred to artificial systems. Machines were therefore

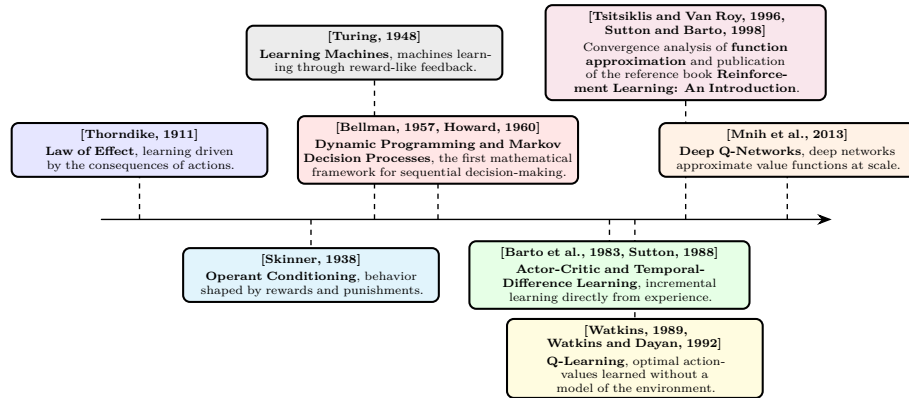


Figure 3.1: Evolution of reinforcement learning prior to its fusion with deep learning.

envisioned as potentially learning through reward-like mechanisms, guided by evaluative feedback signals resembling a “pleasure–pain system” [Turing, 1948]. Although these studies did not involve formal mathematical models, they introduced the central idea of RL, trial-and-error learning guided by environmental feedback.

3.1.2 Sequential decision-making (1950–1970)

A second foundation of RL comes from optimal control theory. This field studies how to determine sequences of decisions that optimize a cumulative performance criterion over time. In this context, the decision-making entity is called an agent. The agent interacts with the environment by selecting actions. The function that maps an observed state to the chosen action is called a policy. The objective is to find an optimal policy, in the sense that it maximizes the expected cumulative reward over time.

This framework led to the formalization of Markov Decision Processes (MDP), which provide a mathematical model for sequential decision-making [Howard, 1960]. MDP describe a decision-maker interacting with an environment through a set of states, actions, rewards, and transition rules. The earliest method proposed to solve MDP is dynamic programming [Bellman, 1957]. It proceeds by breaking down a complex decision problem into smaller, more manageable subproblems, solving each one, and combining the results to obtain an overall solution. However, dynamic programming requires a full and exact description of the environment, which is only feasible for small problems.

3.1.3 Emergence of reinforcement learning (1970–2000)

In order to address this issue, temporal-difference learning methods were introduced [Sutton, 1988]. This mechanism enables incremental learning directly from experience without requiring a complete model of the environment. It can be interpreted as a sample-based approximation of the Bellman equations.

Building on these ideas, the notion of value functions emerged as a way to estimate the long-term benefit of being in a given state or taking a given

action. These functions assign a numerical score to states or state-action pairs, reflecting the expected cumulative reward an agent can obtain from that point onward. They thus provide a compact way to evaluate and compare different courses of action. This line of work culminated in Q-learning, which showed that optimal action-value functions can be learned directly from interaction with the environment, without requiring a model of its dynamics [Watkins, 1989, Watkins and Dayan, 1992].

The publication of the foundational reference work [Sutton and Barto, 1998] contributed to the formalization of RL as a coherent field. From that point onward, the field can be considered well established and widely recognized as a distinct area of research.

3.1.4 Consolidation of reinforcement learning (2000–2013)

A limitation of early algorithms such as Q-learning is their reliance on explicit state-action tables. Although these methods remove the need to know the full dynamics of the MDP, they still require storing a value for each possible state-action pair. As a result, the agent must effectively visit and maintain estimates for a potentially enormous number of states in the environment. This requirement becomes infeasible in large-scale MDP, where the state space is too large to be exhaustively explored or stored in memory.

To address this issue, research shifted toward more general representations of value functions and policies. Instead of maintaining explicit tables, function approximation methods were introduced. These allowed value functions and policies to be represented using parametric models such as linear approximators or perceptron-like architectures [Barto et al., 1983, Tsitsiklis and Van Roy, 1996]. This change enabled generalization across states, reducing the dependence on exhaustive exploration of the state space. The introduction of Deep Q-Networks (DQN), which demonstrated that deep Neural Network (NN) could be used to approximate value functions at scale [Mnih et al., 2013].

Figure 3.1 provides a timeline of the key milestones that shaped the development of reinforcement learning prior to its fusion with deep learning.

3.1.5 Deep reinforcement learning (2013–2026)

The introduction of Deep Learning (DL) into RL marked a major turning point in the field. A series of algorithms significantly improved scalability. Deep Deterministic Policy Gradient extended actor-critic methods to continuous action spaces [Lillicrap et al., 2015]. Proximal Policy Optimization (PPO) later became a standard algorithm in RL due to its simplicity, stability, and ability to perform in both continuous and discrete action spaces [Schulman et al., 2017].

These advances established a practical foundation for large-scale policy optimization. Notably, AlphaZero demonstrated that the combination of deep NN, RL, and tree search could reach superhuman performance in the game of Go [Silver et al., 2017b]. This success was later extended to even more complex real-time strategy environments such as StarCraft II, where agents must handle long-term planning, partial observability, and large action spaces [Samvelyan et al., 2019]. Similarly, in Dota 2, large-scale multi-agent RL systems such as OpenAI Five showed that self-play combined with deep policy

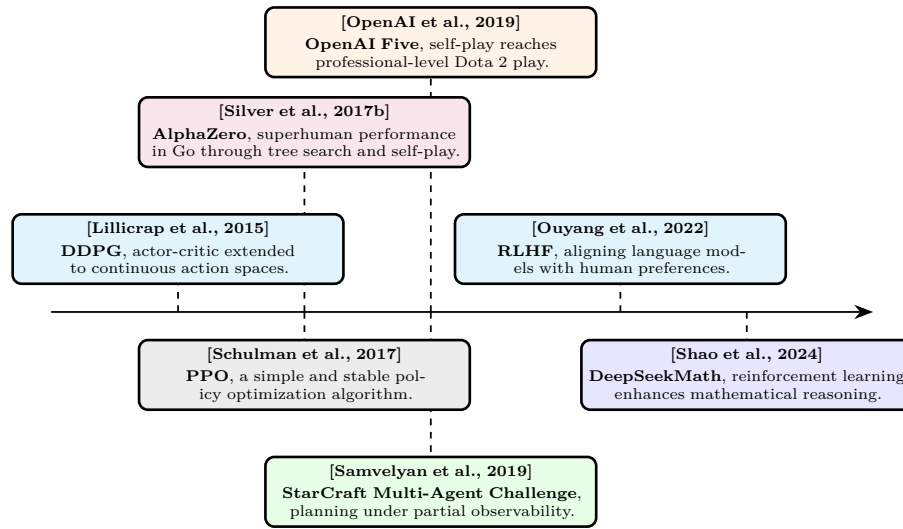


Figure 3.2: Evolution of deep reinforcement learning.

optimization can achieve professional-level performance in highly dynamic and stochastic environments [OpenAI et al., 2019].

More recently, RL has expanded beyond its traditional application domains. With the emergence of increasingly agentic views of Large Language Model (LLM), it has become possible to train these models to perform specific tasks using RL-based techniques. In particular, RL from human feedback has become a method for aligning LLM with human preferences. It does so by formalizing the problem as an MDP and using RL to solve it [Ouyang et al., 2022]. Similarly, DeepSeekMath shows how RL can enhance the mathematical reasoning abilities of LLM [Shao et al., 2024]. It demonstrates that an artificial agent can learn to reason in natural language in order to accomplish a given task. The skills acquired during this learning process also transfer to other similar tasks, leading to improved overall performance. For instance, training a LLM with RL on mathematical problem-solving can also enhance its programming capabilities.

Originally inspired by biological learning processes, RL is now closer than ever to this initial intuition. Agents learn through interaction, adapt via trial and error, and progressively improve their behavior based on experience. This behavior is governed by deep neural architectures, which are the closest computational analog to organic brains. More than ever, RL stands as one of the core paradigms for achieving artificial general intelligence [Silver et al., 2021].

Figure 3.2 provides a timeline of the key milestones that shaped the emergence of deep reinforcement learning.

3.2 Objective

Now that the main developments shaping RL have been outlined, the formal definition of its objective can be introduced. In essence, RL aims to build a system that makes sequential decisions in order to maximize a cumulative reward signal over the long term. The purpose of this section is to translate this intuitive

goal into a rigorous mathematical framework. To this end, the presentation proceeds step by step:

1. Modeling the reinforcement learning problem as a Markov Decision Process.
2. Defining the notion of a policy, which formalizes the agent’s behavior.
3. Describing the interaction between the agent and the environment through trajectories.
4. Introducing value functions to evaluate policies.
5. Defining the notion of an optimal policy.

3.2.1 Markov Decision Processes

The RL framework is grounded in the agent paradigm: the decision-making entity is called the agent. The agent perceives its environment, processes the information it receives, and acts upon the environment to influence future outcomes. To specify the task and the agent’s capacity to perceive and act, the problem is modeled as a MDP.

Before providing a more detailed definition, it is useful to clarify the scope of the MDP problems considered in this manuscript. We therefore restrict the analysis to the following configuration:

- The single-agent learning setting is considered. If any other agents are present in the environment, they are treated as part of the environment dynamics.
- Finite-horizon MDP are considered. Each interaction sequence between the MDP and the policy has a defined beginning and a defined end. Between these two points, the number of interaction steps may vary depending on the situation, but it always remains finite.
- The partially observable framework is adopted, where the agent does not have direct access to the true state of the system, but instead relies on limited observations that provide only partial information about the environment.

The resulting framework can be formally described as a single-agent finite-horizon partially observable Markov decision process. For simplicity, we will refer to this setting as an “MDP” throughout this manuscript.

The following section aims to provide a mathematical description of the RL problem. The formulation adopted in this manuscript has been chosen to emphasize the main elements that will be used throughout the remainder of the manuscript. A MDP is defined by the tuple $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$:

- $\mathcal{S}$: The set of states, representing all possible configurations of the environment. A state belonging to this set is denoted by $s \in \mathcal{S}$. Among these, two special subsets are distinguished: initial states, from which an interaction can begin, and terminal states, in which the interaction ends. The set of initial states is denoted by $\mathcal{S}^i$, with an initial state written as $s^i \in \mathcal{S}^i$. The set of terminal states is denoted by $\mathcal{S}^t$, with a terminal state written as $s^t \in \mathcal{S}^t$.

- 1051 • $\mathcal{O}$: The set of observations that the agent can perceive from the environ-
1052 ment. A observation belonging to this set is denoted by $o \in \mathcal{O}$. Unlike the
1053 true state $s \in \mathcal{S}$, which is hidden from the agent, the observation $o \in \mathcal{O}$
1054 provides partial information about the current state.
- 1055 • $\mathcal{A}$: The set of actions that the agent may execute. A action belonging to
1056 this set is denoted by $a \in \mathcal{A}$. Actions influence the dynamics of the MDP,
1057 thereby shaping its future course. They represent the agent's capacity to
1058 act upon its environment.
- 1059 • $\mathcal{R}$: The set of rewards that the agent can receive from the environment.
1060 A reward belonging to this set is denoted by $r \in \mathcal{R}$. $\mathcal{R}$ is a subset of $\mathbb{R}$.
1061 Rewards can be either positive or negative, depending on whether the
1062 corresponding outcome is considered beneficial or detrimental for the agent.
1063 The agent's goal is to maximize the cumulative sum of these rewards over
1064 time.
- 1065 • p : The transition function, which defines the dynamics of the environment.
1066 Given a state s and an action a , this function defines a probability distribu-
1067 tion over the joint outcome consisting of the next state s' , the observation
1068 o , and the reward r . This function is denoted by p , and is defined as:
1069 $p : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S} \times \mathcal{O} \times \mathcal{R})$. The notation $(s', o, r) \sim p(s, a)$ means that
1070 the tuple (s', o, r) is sampled from the distribution induced by $p(\cdot | s, a)$.
1071 The notation $p(s', o, r | s, a)$ denotes the probability of observing the tuple
1072 (s', o, r) under the distribution $p(\cdot | s, a)$.

1073 The two quantities, observation o and state s , are not independent. They are
1074 jointly generated via $(s', o, r) \sim p(s, a)$, which implicitly defines the relationship
1075 between them. In our MDP formalization, this relationship is left implicit
1076 because it is never manipulated directly during learning.

1077 However, to express mathematical equations, it is convenient to introduce
1078 the conditional distribution over states given an observation. We denote this
1079 distribution by $b : \mathcal{O} \rightarrow \mathcal{P}(\mathcal{S})$. The notation $b(s | o)$ denotes the probability of
1080 state s given the observation o , while $s \sim b(\cdot | o)$ indicates that the state s is
1081 sampled from this distribution.

1082 3.2.2 Policy

1083 The agent's decision-making capability is modeled by a policy function. The
1084 policy defines the stochastic mapping between observations $o \in \mathcal{O}$ and actions
1085 $a \in \mathcal{A}$. A policy function is denoted by π , and is defined as: $\pi : \mathcal{O} \rightarrow \mathcal{P}(\mathcal{A})$. The
1086 notation $a \sim \pi(\cdot | o)$ means that the action a is sampled from the distribution
1087 induced by $\pi(\cdot | o)$. The notation $\pi(a | o)$ denotes the probability of selecting
1088 action a under the distribution $\pi(\cdot | o)$.

1089 It is important to note that this policy does not include any explicit state-
1090 estimation mechanism. Instead, the ability to infer information about the
1091 underlying state is entirely left to the policy itself, which must learn from expe-
1092 rience how to compensate for the missing information. This reflects the design
1093 of many modern RL implementations, where policies are often parameterized
1094 by architectures capable of summarizing past observations, such as recurrent or
1095 attention-based networks, without explicitly modeling the uncertainty over the
1096 underlying state.

1097 3.2.3 Trajectory

1098 Now that both the MDP $(\mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{R}, p)$ and the policy π have been defined,
 1099 the interaction between them can be characterized. A sequence of consecutive
 1100 interactions between an environment and a policy is called a trajectory. A
 1101 trajectory is generated through the repeated interaction between the policy and
 1102 the environment, following the sequence below:

- 1103 1. The environment, being in state s , provides an observation o to the policy
 1104 π . Based on this observation, the policy outputs a probability distribution
 1105 over actions $\pi(\cdot | o)$, from which an action is sampled: $a \sim \pi(\cdot | o)$.
- 1106 2. The sampled action is executed in the environment. The environment then
 1107 generates a transition: $(s', o', r) \sim p(\cdot | s, a)$. It moves to a new state s' ,
 1108 and returns a new observation o' and a reward r .

1109 A trajectory t of length T can be written as:

$$t = (s_1, o_1, a_1, r_1, s_2, o_2, a_2, r_2, \dots, s_T, o_T, a_T, r_T).$$

1110 It is important to clearly distinguish between a trajectory distribution and its
 1111 realizations. The distribution over trajectories is denoted by τ , and a particular
 1112 trajectory is denoted by t . The distribution τ is determined by the initial state
 1113 s , the initial observation o , the environment dynamics p , and the policy π , and is
 1114 denoted by $\tau(s, o, p, \pi)$. The notation $t \sim \tau(s, o, p, \pi)$ means that the trajectory
 1115 t is sampled from the distribution induced by s , o , p , and π . The notation
 1116 $\tau(t | s, o, p, \pi)$ denotes the probability of observing the trajectory t under the
 1117 distribution $\tau(\cdot | s, o, p, \pi)$.

1118 An important property of the trajectory distribution is its recursive structure.
 1119 Since both the policy π and the transition function p are known, the distribution
 1120 over trajectories starting from s_1 with observation o_1 can be expressed in terms of
 1121 the distribution over trajectories starting from the next state s_2 with observation
 1122 o_2 . Specifically, the probability of a trajectory t factorizes as:

$$\tau(t | s_1, o_1, p, \pi) = \pi(a_1 | o_1) p(s_2, o_2, r_1 | s_1, a_1) \tau(t_2 | s_2, o_2, p, \pi),$$

1123 where $t_2 = (s_2, o_2, a_2, r_2, \dots, s_T, o_T, a_T, r_T)$ denotes the trajectory starting from
 1124 the second step.

1125 There exist two special types of trajectories. First, a trajectory that ter-
 1126 minates in a terminal state is called a terminal trajectory, and is denoted
 1127 by $t_t \sim \tau_t(s, o, p, \pi)$. Second, a terminal trajectory that starts from an ini-
 1128 tial state and an initial observation is called an episode, and is denoted by
 1129 $t_e \sim \tau_t(s^i, o, p, \pi)$.

1130 3.2.4 Value Functions

1131 To evaluate the ability of a policy π to collect rewards, the notion of return is
 1132 introduced. The return of a trajectory $g(t, \gamma)$, is defined as the discounted sum
 1133 of rewards collected along the trajectory:

$$g(t, \gamma) = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots + \gamma^{T-1} r_T = \sum_{i=1}^T \gamma^{i-1} r_i,$$

where $\gamma \in [0, 1]$ is the discount factor. This parameter controls the time horizon of the agent's objective. When γ is close to 0, the agent focuses almost exclusively on immediate rewards, leading to short-sighted behavior. As γ increases toward 1, future rewards gain more weight, and the agent adopts a more far-sighted strategy. In theory, since the goal is to maximize long-term cumulative reward, the discount factor would ideally be set to $\gamma = 1$. In practice, however, γ is often chosen slightly below 1 (e.g., 0.99) to stabilize learning, and to reflect the fact that distant future rewards are inherently more uncertain.

The return also admits a recursive form, which is central to the dynamic programming and RL algorithms presented later:

$$g(t, \gamma) = r_1 + \gamma g(t_2, \gamma),$$

From the notion of return, functions that estimate the expected return of a policy from a given starting point can be defined. These functions are called value functions. The first one considered is the state-value function V . It tells us how desirable a particular state is under a given policy π , given an observation o , the environment dynamics p , and the discount factor γ . Strictly speaking, this function should be written as $V(s, o, p, \gamma, \pi)$ to reflect all its dependencies. However, since both the environment dynamics p and the discount factor γ are fixed throughout learning, they are omitted from the notation. To emphasize that the value is tied to a specific policy, π is placed as a subscript. This yields the compact notation $V_\pi : \mathcal{S} \times \mathcal{O} \rightarrow \mathbb{R}$.

If $V_\pi(s, o) > V_\pi(s', o')$, then, under policy π , the agent prefers being in state s with observation o over state s' with observation o' . Formally, the state-value function is defined as:

$$V_\pi(s, o) = \mathbb{E}[g(t, \gamma)], \quad \text{with } t \sim \tau_t(s, o, p, \pi).$$

It is worth noting that the state-value function V_π is recursive, a property that follows directly from the recursive form of the return:

$$\begin{aligned} V_\pi(s, o) &= \mathbb{E}[g(t, \gamma)], \quad \text{with } t \sim \tau_t(s, o, p, \pi) \\ &= \mathbb{E}[r_1 + \gamma g(t_2, \gamma)] \\ &= \mathbb{E}[r_1 + \gamma V_\pi(s_2)] \end{aligned}$$

If the expectation is expressed explicitly in terms of the distributions p and π , the Bellman equation for V_π is obtained:

$$\begin{aligned} V_\pi(s, o) &= \mathbb{E}[r_1 + \gamma V_\pi(s_2)] \\ &= \sum_{a \in \mathcal{A}} \pi(a | o) \sum_{\substack{s' \in \mathcal{S} \\ o' \in \mathcal{O} \\ r \in \mathcal{R}}} p(s', o', r | s, a) [r + \gamma V_\pi(s', o')]. \end{aligned}$$

Now that the value of a state has been expressed, the next step is to consider the value of taking a specific action in a given state. Indeed, it is useful to estimate the expected future gain of an action a in state s , in order to compare

different actions and determine which one is the most promising under a given policy π . This function is called the state-action value function $Q_\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. If $Q_\pi(s, a) > Q_\pi(s, a')$, then, under policy π , choosing action a in state s yields a higher expected return than choosing action a' . It is defined as:

$$Q_\pi(s, a) = \mathbb{E} \left[r + \gamma g(t, \gamma) \right], \quad \text{with } t \sim \tau_t(s', o, p, \pi), \quad (s', o, r) \sim p(\cdot \mid s, a).$$

The function Q_π can be related to V_π by recognizing that, after taking action a in state s , the remaining return from the next state s' is precisely $V_\pi(s')$. This yields:

$$\begin{aligned} Q_\pi(s, a) &= \mathbb{E} \left[r + \gamma g(t, \gamma) \right], \quad \text{with } t \sim \tau_t(s', o, p, \pi), \quad (s', o, r) \sim p(\cdot \mid s, a) \\ &= \mathbb{E} \left[r + \gamma V_\pi(s', o) \right] \\ &= \sum_{\substack{s' \in \mathcal{S} \\ o' \in \mathcal{O} \\ r \in \mathcal{R}}} p(s', o', r \mid s, a) \left[r + \gamma V_\pi(s', o) \right] \end{aligned}$$

3.2.5 Optimal Policy

The goal of RL is to find a policy that maximizes the expected cumulative reward from any observation until the end of the trajectory. This policy is called the optimal policy and is denoted by π^* . In a partially observable setting, the optimal action is the one that maximizes the expected state-action value over the distribution of possible states given the observation:

$$\pi^*(a \mid o) = \begin{cases} 1 & \text{if } a = \arg \max_{a' \in \mathcal{A}} \sum_{s \in \mathcal{S}} b(s \mid o) Q_{\pi^*}(s, a'), \\ 0 & \text{otherwise} \end{cases}$$

As shown, both V_π and Q_π can be expressed explicitly in terms of the environment dynamics p and the policy π . If p , b and π are fully known, one can theoretically compute the optimal policy by unfolding the graph of all possible trajectories starting from a given state s and observation o . This is possible precisely because V_π and Q_π are recursive functions. Solving an MDP in this way is known as dynamic programming.

In practice, most environments have too many states to allow an explicit representation of the dynamics p . Storing and computing over the full transition function becomes intractable. To overcome this limitation, instead of directly constructing the optimal policy, an iterative approach is adopted: starting from an initial policy, it is progressively improved so that it converges toward π^* . This iterative improvement through interaction with the environment is precisely what constitutes learning in RL, and it is the subject of the next section.

3.3 Learning

Now that the optimal policy π^* has been defined, the practical question of how to obtain it arises. The recursive Bellman equations for V_π and Q_π suggest

dynamic programming approaches, but these scale poorly to large state spaces. RL offers an alternative: it seeks to solve the MDP iteratively, by generating a sequence of policies that progressively approach π^* through direct interaction with the environment. In this section, two major families of RL methods used today are introduced: critic-only methods and actor-critic methods.

3.3.1 Critic-Only Methods

The central idea behind all critic-only approaches is to iteratively construct an approximation of the state-action value function Q_π , denoted by $\hat{Q}_\pi$. Although $\hat{Q}_\pi$ is only an approximation of the true Q_π , the policy always acts greedily with respect to this approximation:

$$\pi(a \mid o) = \begin{cases} 1 & \text{if } a = \arg \max_{a' \in \mathcal{A}} \hat{Q}_\pi(o, a'), \\ 0 & \text{otherwise.} \end{cases}$$

Since $\hat{Q}_\pi$ serves as the basis for the agent's decisions, it cannot rely on more information than the agent itself has access to. It must therefore be defined solely over observations, as a function $\hat{Q}_\pi : \mathcal{O} \times \mathcal{A} \rightarrow \mathbb{R}$. Its objective is to approximate the true Q_π , which depends on the underlying states s , using only the partial information available through o . Once again, this poses no issue in practice, as it reflects how these algorithms are actually deployed in real-world applications.

It is precisely this way of modeling the policy that gives this family of methods its name. The policy is not represented explicitly as a standalone function. Rather, it simply emerges from searching over the set of actions to maximize $\hat{Q}_\pi$.

To give a clear example of critic-only methods, they are presented in their simplest form: model-free, without temporal-difference learning, without replay buffer, without bootstrapping, and using only ε -greedy exploration. Naturally, many variants and extensions of this basic scheme exist, but the core idea remains the same.

When implementing a RL algorithm, one of the first questions is how to represent the value function. In critic-only methods, this means choosing a model for $\hat{Q}_\pi$. Any machine learning model capable of regression task can be used for this purpose. The set of parameters of the model is denoted by θ . In deep RL, a NN is used to model this function, in which case θ represents the weights and biases of the network.

In our simple setting, exploration is handled via ε -greedy selection. With probability $1 - \varepsilon$, the policy selects the greedy action that maximizes $\hat{Q}_\pi$, and with probability ε it selects an action uniformly at random. The trajectory distribution induced by the resulting policy now depends on both π and ε , and is denoted by $t \sim \tau(s, o, p, \pi, \varepsilon)$. The parameter ε controls the degree of exploration. If the policy always chooses the action it currently believes to be optimal, it may miss better alternatives that $\hat{Q}_\pi$ has not yet learned to value correctly. Exploring suboptimal actions from time to time is therefore essential to verify that $\hat{Q}_\pi$ accurately reflects the true values, while still focusing most of the time on the most promising actions. In more advanced implementations, ε can be adapted over the course of learning to control exploration dynamically, for instance by decreasing it gradually over time.

1237 The learning process consists of two steps that repeat until convergence:

- 1238 1. **Data collection.** The current policy π interacts with the MDP. For
 1239 a given starting state s_1^i and observation o_1 , the interaction produces a
 1240 episode t_e sampled from $\tau_t(s_1^i, o_1, p, \pi, \varepsilon)$. For each step i along the episode,
 1241 the return from that step onward is $g(t_{i:}) = \sum_{k=i}^T \gamma^{k-i} r_k$, which serves as
 1242 the target value for the state-action pair (s_i, a_i) .
- 1243 2. **Model update.** The tuples $(s_i, a_i, g(t_{i:}))$ obtained in the previous step
 1244 serve as supervised training data: the state-action pair (s_i, a_i) is the input,
 1245 and the return $g(t_{i:})$ is the associated target. These examples are used
 1246 to train $\hat{Q}_\theta$. The improved approximation is denoted by $\hat{Q}'_\theta$, which now
 1247 predicts returns more accurately under π . Since $\hat{Q}'_\theta$ has changed, the policy
 1248 derived from it also changes. This new policy is denoted by π' .

1249 Because the policy is now π' , the approximation $\hat{Q}'_\theta$ must be trained to predict
 1250 returns under π' . This requires returning to step 1 to collect fresh interaction
 1251 data generated by π' . The two steps thus alternate until the policy and the
 1252 value approximation stabilize.

1253 Even this simple algorithm already illustrates a difficulty inherent to all forms
 1254 of RL: the instability of the learning process. Each update from $\hat{Q}_\theta$ to $\hat{Q}'_\theta$ causes
 1255 the policy to shift from π to π' . For the learning to remain stable, the change
 1256 from $\hat{Q}_\theta$ to $\hat{Q}'_\theta$ should be as small as possible. This way, $\hat{Q}'_\theta$, trained to predict
 1257 returns under π , remains reasonably accurate under π' . However, the smaller
 1258 the change, the slower the learning progresses. A trade-off must therefore be
 1259 struck between stability and learning speed. Many improvements introduced
 1260 later in the literature aim at stabilizing this learning loop [Hessel et al., 2017].

1261 Convergence guarantees for such algorithms have been proved under a different
 1262 set of assumptions than the setting presented here. For example, in the classical
 1263 convergence results, the MDP is fully observable, the value function is stored
 1264 in a table, and the learning rate is decreased at a suitable rate over time
 1265 [Watkins and Dayan, 1992]. In modern implementations, the value function is
 1266 approximated by a NN. Theoretical convergence proofs no longer hold in this
 1267 setting. Nevertheless, deep methods have demonstrated remarkable empirical
 1268 success, making them the tool of choice for RL tasks.

1269 Critic-only approaches suffer from several limitations. The main limitation
 1270 is scalability to large action spaces. Since the policy must evaluate all possible
 1271 actions at each decision step to select the maximum, the computational cost
 1272 grows with the size of the action set. To address this issue, another family of
 1273 RL algorithms exists: actor-critic methods.

1274 3.3.2 Actor-Critic Methods

1275 Two principal modifications distinguish actor-critic methods from critic-only
 1276 approaches. First, the objective is to approximate the state-value function V ,
 1277 instead of the state-action value function Q . Second, the policy π is given its
 1278 own dedicated set of parameters. Two separate components are thus obtained:
 1279 a value function approximation $\hat{V}_{\theta_1}$ and a parameterized policy π_{θ_2} .

1280 Analogously to critic-only methods, $\hat{V}_{\theta_1}$ is trained using supervised learning
 1281 on pairs obtained from interaction with the MDP. For each step i along a
 1282 trajectory, the input is the state-observation pair (s_i, o_i) and the target is the

1283 observed return $g(t_{i:})$. The value function is thus trained to correct its predictions
 1284 based on data collected by the current policy interacting with the environment.

1285 The key difference lies in how the policy π itself is optimized. Since the policy
 1286 now has its own set of parameters θ_2 , it can be updated directly. The update
 1287 uses the notion of advantage $A : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{R}$, defined as:

$$A(s, a) = g(t) - \hat{V}_{\theta_1}(s, o),$$

1288 where t is the trajectory that starts by taking action a in state s and then follows
 1289 π for all subsequent steps.

1290 When $A(s, a) > 0$, the action a performed better than expected, and the
 1291 policy parameters θ_2 are adjusted to increase the probability $\pi_{\theta_2}(a | o)$. When
 1292 $A(s, a) < 0$, the action a performed worse than expected, and the probability is
 1293 decreased.

1294 Both $\hat{V}_{\theta_1}$ and π_{θ_2} are improved simultaneously within the same loop: the
 1295 critic provides a learned baseline for the actor, and the actor generates the data
 1296 used to train the critic. This interplay reduces the instability inherent to RL.

1297 PPO (Proximal Policy Optimization), which is the current standard in RL,
 1298 belongs to this family of actor-critic methods. It will be used throughout this
 1299 manuscript for all RL experiments.

1300 Now that highly performant policies can be trained, learning purely through
 1301 interaction with their environment without any human supervision, the question
 1302 arises of how to explain their behavior. This is the subject of the next chapter:
 1303 explainability in the context of RL.

Chapter 4

Explainability in Reinforcement Learning

Today, the majority of Artificial Intelligence (AI) agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences [Silver et al., 2017a]. Even more autonomous systems, such as agents based on Large Language Models (LLMs), remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval [Chen et al., 2021, Lewis et al., 2021]. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees [Seshia et al., 2022]. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both Reinforcement Learning (RL) and Deep Learning (DL) [Mnih et al., 2013]. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: eXplainable Reinforcement Learning (XRL), a subfield of eXplainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of RL agents.

This chapter presents the main existing XRL methods. We first define what is meant by an explanation in the context of XAI. We then propose a taxonomy for organizing existing XRL methods. This taxonomy serves as the basis for the state-of-the-art review presented in the remainder of the chapter.

1333 4.1 Explanation in Artificial Intelligence

1334 In this section, we define what an explanation is in the context of XAI and how
 1335 it can be obtained. We first introduce the concept of explanation in general,
 1336 along with the explainability method used to produce it. We then describe
 1337 how this concept applies to XAI. Finally, we further detail the properties that
 1338 characterize an explainability method in the context of XAI.

1339 4.1.1 Explanation

1340 Establishing what is meant by an explanation is challenging due to the funda-
 1341 mentally multidisciplinary nature of the concept. When we refer to explanation,
 1342 we are inherently referring to an explanation provided to a human. As soon as
 1343 humans are involved, the problem is no longer purely computational or mathe-
 1344 matical. It involves several fields of research, such as philosophy, psychology, and
 1345 sociology. So far, these disciplines have not converged on a unified definition of ex-
 1346 planation that can be directly applied to XAI [Miller, 2019, Bellucci et al., 2021].

1347 For this reason, we propose the following definition of explanation. An
 1348 explanation involves two entities: the object (Definition 4.1.1) being explained
 1349 and the receiver of the explanation. The receiver is a human who aims to
 1350 understand how the object operates and will be referred to as the observer
 1351 throughout this manuscript (Definition 4.1.2). The observer seeks to understand
 1352 the object and, in doing so, builds a mental model of its functioning. An
 1353 explanation is therefore what allows the observer to refine and correct their
 1354 mental model of how the object operates (Definition 4.1.4). This refinement is
 1355 obtained through a transformation process that produces an explanation from
 1356 the object being explained. We refer to this process as an explainability method
 1357 (Definition 4.1.5), detailed further in Section 4.1.3.

1358 For an observer to understand an explanation, the information it contains
 1359 must be interpretable (Definition 4.1.3). Information is considered interpretable
 1360 when it can be directly understood by a human without requiring any additional
 1361 transformation process.

Definition 4.1.1: Object

The object of an explanation is the system the observer wants to understand.

1362

Definition 4.1.2: Observer

The observer is the human who aims to understand an object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

1363

Definition 4.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

1364

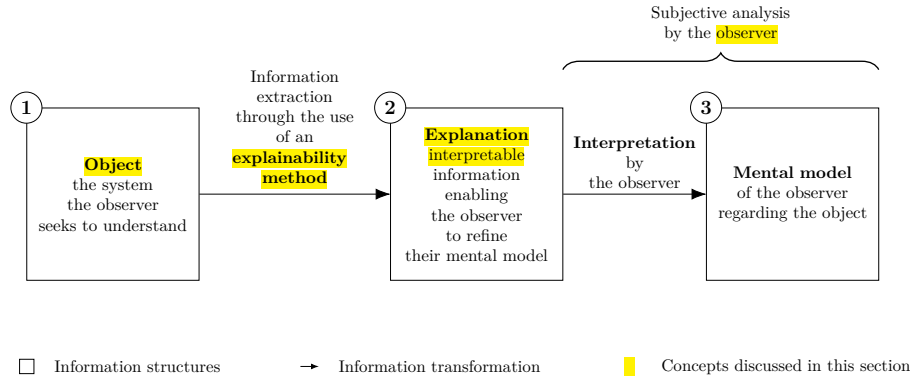


Figure 4.1: Explanation process.

Definition 4.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

Definition 4.1.5: Explainability Method

An explainability method is a transformation process that produces an explanation from the object being explained.

This definition of explanation provides the foundation for the approach to explainability adopted throughout this manuscript. This work adopts the perspective of mechanistic explanation in the philosophy of science [Siegel and Craver, 2024, Zednik, 2019]. Within this framework, explainability is not regarded as a binary property but as a continuum that evolves with the development of scientific knowledge.

Accordingly, the opacity of a system should be interpreted as an epistemic limitation reflecting the current state of knowledge rather than as an intrinsic property of the system itself. This view is consistent with the methodological commitments of scientific inquiry. Scientific progress is driven by the continuous refinement of theories and analytical methods that provide progressively deeper mechanistic accounts of observed phenomena. Consequently, explainability should be regarded as a matter of degree, increasing as these mechanisms become better understood.

This quantitative perspective allows us to express an intuitive idea: not all explanations provide the same level of value [Doshi-Velez and Kim, 2017]. The effectiveness of an explanation depends on its ability to improve the observer's understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer's misunderstanding of a key mechanism.

Figure 4.1 summarizes this whole explanation process, from the object of the explanation to the mental model refined by the observer, as a chain of three information structures, depicted as boxes and connected by arrows denoting a transformation of information:

- 1391 (1) The **object** of the explanation, that is, the system the observer seeks to
1392 understand.
- 1393 (2) The **explanation**, the interpretable information produced from the object.
1394 This transformation is performed through the use of an explainability
1395 method.
- 1396 (3) The **mental model** held by the observer regarding the object.
- 1397 A brace spanning boxes (2) and (3) indicates that this second half of the process,
1398 from the explanation to the resulting mental model, constitutes a subjective
1399 analysis carried out by the observer.

1400 4.1.2 Explainable Artificial Intelligence

1401 Now that the general framework of explanation has been established, we can exam-
1402 ine how this concept applies in the context of AI. In the context of XAI, the object
1403 of explanation is an AI system. More precisely, XAI emerged as a response to the
1404 success of DL models [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In
1405 these architectures, information is represented in a distributed manner across the
1406 network through patterns of activation and connections between computational
1407 units. Unlike symbolic approaches, individual components of Neural Networks
1408 (NNs) cannot be directly associated with human-defined concepts. From this
1409 starting point, we identify two types of approaches:

- 1410 • **Substitutive explainability:** In this approach, the underlying assump-
1411 tion is that there is nothing to explain within a NN. There is no semantic
1412 meaning associated with each computational component, and therefore
1413 nothing to interpret. This corresponds to the so-called “black box” view
1414 of NNs. Under this assumption, XAI consists of replacing as much of the
1415 DL system as possible with alternative AI methods [Rudin, 2019].
- 1416 • **Intrinsic explainability:** In this approach, it is argued that NNs in-
1417 herently contain mechanisms that can be explained. The limitation is
1418 considered to lie not in the networks themselves, but in the absence of
1419 adequate tools to analyze their functioning [Olah et al., 2020].

1420 This manuscript considers methods from both approaches in order to provide
1421 a comprehensive overview of the XAI field. According to the definition of
1422 explanation adopted in this manuscript (Definition 4.1.4), we consider that any
1423 system can be explained. As a consequence, the perspective adopted throughout
1424 this manuscript is that of intrinsic explainability.

1425 4.1.3 Explainability Method in XAI

1426 Now that the general framework of XAI has been established, we need to further
1427 detail the properties of an explanation in this context. In our framework, an
1428 explanation in the field of XAI can be characterized by three components:

- 1429 • **Source:** refers to the origin of the information used by the explainability
1430 method. The explainability method transforms the raw information pro-
1431 duced by the AI system into an explanation that can be understood by an
1432 observer.

1433 • **Form:** refers to the representation through which the explanation is
 1434 communicated. The form defines how the information is presented to
 1435 the observer and must allow human interpretation. An explanation can
 1436 take many different forms, including text, images, computer code, and
 1437 mathematical equations.

1438 • **Subject:** refers to the aspect of the AI system that is being explained. In
 1439 XAI, this distinction is commonly associated with the difference between
 1440 local and global explanations. A local explanation describes why an AI
 1441 system produced a specific output in a given situation. A global explanation
 1442 aims to describe the general functioning of the AI system.

1443 These three properties of an explanation are entirely determined by the
 1444 explainability method (Definition 4.1.5) used to produce it. In this context,
 1445 an explainability method is a transformation process that maps a source of
 1446 information into an explanation characterized by a specific form and subject.
 1447 Research in XAI focuses on developing new methods capable of performing this
 1448 transformation in order to overcome the lack of interpretability associated with
 1449 DL models.

1450 4.2 Taxonomy of XRL Methods

1451 Now that the concept of XAI has been defined, we turn to its application in RL,
 1452 known as XRL. As in XAI, explanations in XRL are produced by explainability
 1453 methods and can be described in terms of their subject, source, and form. This
 1454 section reviews the possible variations of these three aspects in the specific
 1455 context of RL.

1456 4.2.1 Subject

1457 The first question to address is the subject of explanation in the context of RL.
 1458 To answer this question, it is necessary to identify the different components
 1459 involved in a RL system and determine which of them constitutes the object of
 1460 explanation. A RL system can be described through two main components: the
 1461 environment and the policy.

1462 The environment defines the dynamics in which the agent evolves, including
 1463 the possible states, observations, actions, and transitions. Environments are
 1464 simulations designed by humans, and their dynamics are explicitly defined.
 1465 Therefore, it is assumed that the environment is known to the observer and does
 1466 not require explanation.

1467 The main object of interest is therefore the agent’s policy. The policy
 1468 represents the decision-making mechanism of the agent. It is a function that maps
 1469 observations to actions and is learned through interaction with the environment in
 1470 order to maximize the expected cumulative reward. The policy is the component
 1471 that encodes the learning process and that XRL aims to explain.

1472 Explaining a policy introduces additional challenges compared with classical
 1473 XAI settings. In traditional XAI problems, the objective is to explain an isolated
 1474 decision, such as determining which features of an image led to a classification
 1475 result. In contrast, RL introduces a temporal dimension into the decision-making
 1476 process. An action is not only determined by the current observation but also

1477 influences future states and rewards. Therefore, understanding the behavior of
 1478 an agent may require considering a sequence of interactions between the agent
 1479 and its environment.

1480 This temporal dimension leads to a distinction between two objectives of
 1481 explanation in XRL:

- 1482 • **Decision explanations:** focus on identifying the elements of the current
 1483 observation that influenced the action selected by the policy. This type of
 1484 explanation is close to classical XAI approaches, as it analyzes the rela-
 1485 tionship between inputs and outputs while abstracting away the sequential
 1486 nature of the decision process.
- 1487 • **Strategy explanations:** aim to characterize the agent’s behavior over
 1488 time. They focus on sequences of decisions and interactions with the
 1489 environment in order to understand how the agent achieves its long-term
 1490 objective. This type of explanation is specific to RL, as it addresses the
 1491 temporal and goal-oriented nature of the learning process.

1492 These two objectives should not be considered as strictly separated categories,
 1493 but rather as two extremes of a continuum. Indeed, understanding the elements
 1494 of an observation that influence a specific decision can help the observer generalize
 1495 this information to infer broader aspects of the agent’s behavior. Conversely,
 1496 understanding the agent’s overall strategy can provide insights into the reasoning
 1497 behind individual decisions.

1498 4.2.2 Source

1499 We identify three sources of explanation in RL, all obtained during or after the
 1500 learning process and containing information about the agent:

- 1501 • **Policies:** policies are functions that generate a distribution over the action
 1502 space for a given observation. The objective of RL is to learn a policy that
 1503 maximizes the expected cumulative future rewards. During the learning
 1504 phase, the policy is iteratively improved through interactions with the
 1505 environment.
- 1506 • **Trajectories:** trajectories represent sequences of successive interactions
 1507 between the agent and the environment. At each step, the agent selects
 1508 an action based on its observation, which modifies the environment and
 1509 produces a reward as well as a new observation. This process continues
 1510 until the end of the trajectory.
- 1511 • **Value functions:** value functions provide predictions about the future
 1512 outcome of a trajectory from a given observation. The most commonly
 1513 used value function is the critic function, which estimates the expected
 1514 sum of future rewards. This prediction guides the learning process, as the
 1515 agent updates its policy to maximize the estimated future return. However,
 1516 other types of value functions can also be used as sources of explainability.

1517 4.2.3 Form

1518 We identify six forms of explanation in the RL setting:

Explanation Subject	Explanation Form	Explainability Method Family	Source of Explanation		
			Policy	Trajectories	Value functions
Decision	Policy Model	<i>Interpretable policy learning</i>	✓		
		<i>Policy approximation via interpretable model</i>	✓		
	Saliency Map	<i>Observation perturbation</i>	✓		
		<i>Observation gradient analysis</i>	✓		
		<i>Attention-based architecture</i>	✓		
	Counterfactual	<i>Latent space usage for counterfactual generation</i>	✓		
Strategy	Agent Trajectory	<i>Hierarchical decomposition</i>	✓	✓	
		<i>Extraction via critic analysis</i>		✓	✓
	Trajectory Prediction	<i>Critic enrichment</i>			✓
	Interaction Model	<i>Bayesian network learning</i>		✓	
		<i>Latent space usage for MDP generation</i>	✓		

Table 4.1: Taxonomy of explainability methods in reinforcement learning.

- **Policy Model:** an interpretable model of the agent’s policy function.
- **Saliency Map:** a weighting of the input features representing their influence on the output of the policy function.
- **Counterfactual:** a modified observation obtained by perturbing the original observation in order to produce a different action from the agent.
- **Agent Trajectory:** a sequence of successive interactions between the agent and the environment.
- **Trajectory Prediction:** a prediction about the future evolution of the trajectory produced by a value function.
- **Interaction Model:** an interpretable model of the interactions between the agent and the environment.

Among the three components characterizing an explanation, the form is the one that varies most significantly across explainability methods. While subjects and sources tend to be relatively stable in the XRL literature, the diversity in how explanations are presented is considerably greater. The list above is by no means definitive and may evolve as new approaches emerge. For this reason, the following section is organized according to the form of explanations and presents existing explainability methods based on this criterion.

4.3 State-of-the-Art Review of XRL Methods

The purpose of this section is to detail the functioning of existing explainability methods (see Table 4.1). They are presented according to the form of explanation they produce, following the taxonomy introduced in Section 4.2.

Each subsection introduces one of the six forms of explanation identified in Section 4.2.3. Within each subsection, each subsubsection is dedicated to a family of methods capable of producing that form of explanation.

A family of methods is defined as a set of methods that share the same source, form, and subject of explanation, as well as a similar procedure for generating explanations. Methods belonging to the same family may differ in their implementation or in the details of the generation process, while relying on the same underlying principle.

4.3.1 Policy Model

The policy function can be represented by a wide variety of function classes. In modern RL, NNs is the most commonly used models for representing policies. Although these models achieve the highest performance, their connectionist nature makes their internal representations non-interpretable.

For this reason, a substitutive approach to XRL aims to replace these models with alternative models considered interpretable. Such models can be understood through direct analysis of their representations. The main classes of interpretable models used to represent policy functions include decision trees [Topin et al., 2021], algorithms [Trivedi et al., 2022], formal logic [Payani and Fekri, 2019], and linear models [Garreau and von Luxburg, 2020].

Explanations derived from an interpretable model primarily provide insights into the decision-making process, since what is modeled is the mapping between an observation and an action. If the model is sufficiently simple or well structured, the observer can form a mental representation of a sequence of decisions across the environment and thus obtain information about the agent’s strategy.

The main limitation of these methods is that interpretable models rely on symbolic representations. When the policy to be represented is too complex, achieving comparable performance requires a large number of rules or conditions. The resulting model can quickly become difficult to analyze and lose its interpretability.

For example, representing a policy capable of playing Go using a symbolic model would require considering a very large number of possible situations and specific cases. Even if such a representation were possible, the resulting model would likely become intractable for human analysis due to its complexity. This illustrates why NN-based approaches have become dominant in such domains, as they can learn complex decision strategies without requiring an explicit enumeration of rules.

To obtain an interpretable policy model, two families of explainability methods are available: *interpretable policy learning* and *policy approximation via interpretable model*.

Interpretable Policy Learning

This family of methods aims at learning a policy represented by an **interpretable** model [Topin et al., 2021, Trivedi et al., 2022]. It is the most direct approach to addressing the challenges posed by XRL. This type of training requires modifying the Markov Decision Processes (MDP) in order to adapt it to the search space and to the type of policy representation. Moreover, this search space often needs to be reduced to make learning feasible. Such a **reduction** requires the

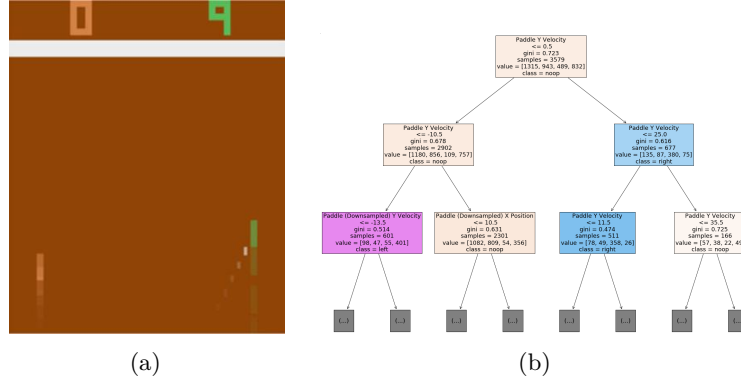


Figure 4.2: Policy approximation via an interpretable model [Sieusahai and Guzdial, 2021]. (a) shows an observation from the Atari game “Pong”, and (b) shows the decision tree obtained by approximating a deep reinforcement learning policy trained on this environment.

integration of **expert knowledge** into the learning process. With this family of methods, the *interpretable policy function* itself is used to provide explanations to the observer.

Policy Approximation via Interpretable Model

An interpretable model can be trained to approximate a policy learned through deep RL [Sieusahai and Guzdial, 2021]. In this setting, the problem is formulated as a supervised learning task, where the original deep RL policy is used to label each observation with the action it selects (see Figure 4.2a). Using this dataset, the interpretable model is trained to reproduce the behavior of the original policy, yielding for instance the decision tree illustrated in Figure 4.2b. The resulting interpretable model is then used as an explanation for the observer. If this approximation is sufficiently faithful, it can also be deployed in the environment in place of the original policy, thereby providing a high degree of control over the agent.

4.3.2 Saliency Map

The influence of each part of the input on the action selected by the policy can be visualized using heatmaps (Figure 4.3). In the RL setting, the input corresponds to the observation received by the agent at a given time step. Heatmaps therefore highlight the regions of the observation that have the greatest influence on the action selection process. By providing this information, they allow the observer to identify which elements of the environment are considered important by the policy when making a decision.

A limitation of heatmap-based methods is that the resulting explanations are not always directly meaningful to the observer. As illustrated in Figure 4.3b, interpreting which regions are important often requires human reasoning and involves a degree of subjectivity. This limitation becomes particularly relevant in RL, where the objective is not only to explain a single decision but also to understand the factors influencing a sequence of decisions over time.

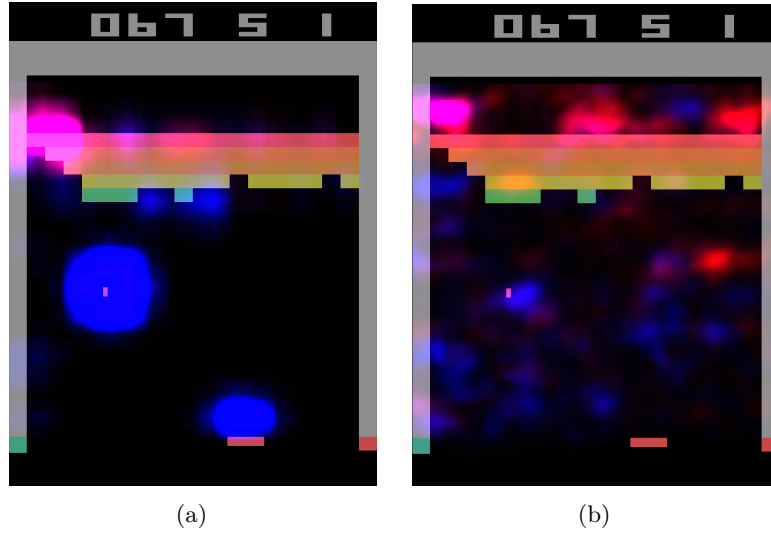


Figure 4.3: Heatmaps obtained by [Greydanus et al., 2018] for the same observation of the Atari game “Breakout”. Highlighted areas correspond to the most influential regions for the policy function’s decision. (a) is obtained via observation perturbation (see Section 4.3.2), whereas (b) is obtained via observation gradient analysis (see Section 4.3.2).

Three families of explainability methods can be used to obtain this type of explanation: *observation perturbation*, *observation gradient analysis*, and *attention-based architectures*.

Observation Perturbation

Perturbation analysis consists in applying various **modifications** to the observation in order to study their **impact** on the output of the **policy function** [Greydanus et al., 2018]. The greater the variation in the output caused by a perturbation applied to a given component of the input vector, the higher the **weight** assigned to that component. The magnitude of the perturbation is measured by computing the distance between the original output and the output obtained from the perturbed input. Using this set of weights, a *heatmap* is constructed, enabling the observer to visualize which parts of the observation vector are most sensitive in the action-selection process (see Figure 4.3a).

Observation Gradient Analysis

Observation **gradient** analysis refers to a set of methods that require a differentiable policy function in order to extract information from it [Simonyan et al., 2014, Weitekamp et al., 2019, Huber et al., 2019]. To this end, the gradient of the policy function is computed with respect to all **components of the input vector**. For each input component, the resulting gradient provides a weighting that represents its importance in the output of the policy function. Based on these weightings, a *heatmap* is constructed to allow the observer to visually identify the parts of the input vector that are the most influential in the policy function’s

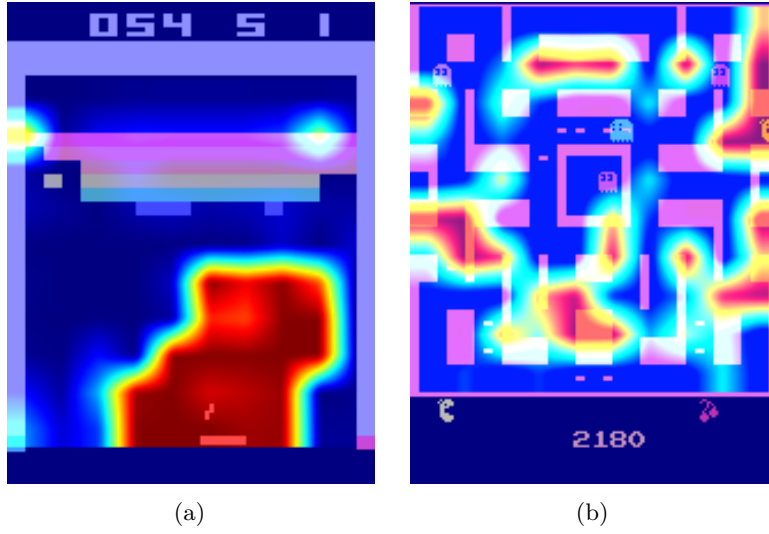


Figure 4.4: Attention-based heatmaps obtained by [Mott et al., 2019] for the Atari games “Breakout” and “Ms. Pac-Man” (see Section 4.3.2). In (a), the highlighted regions are easily interpretable, whereas in (b), the attention is spread across many regions of the observation, making the explanation much harder to interpret.

1637 output (see Figure 4.3b).

1638 *Attention-Based Architecture*

1639 **Attention** blocks constitute a NN architecture that makes it possible to weight
 1640 the relative importance of one part of a vector with respect to the other parts
 1641 of the same vector [Vaswani et al., 2023]. These weights are computed as a
 1642 function of the vector itself and are learned by the NN during training. For
 1643 a given observation, each component of the observation vector is assigned a
 1644 scalar value. It can therefore be assumed that the components associated
 1645 with the highest weights are the most influential in determining the output
 1646 [Mott et al., 2019, Itaya et al., 2021]. This attention-based *heatmap* is then
 1647 provided to the observer as the explanation (see Figure 4.4a). However, as
 1648 shown in Figure 4.4b, the attention weights are not always concentrated on a
 1649 small, easily interpretable set of regions, which can make this type of explanation
 1650 difficult to exploit.

1651 4.3.3 Counterfactual

1652 In the context of RL, counterfactual explanations are used to generate new
 1653 observations. This new observation, distinct from the original one, leads the
 1654 agent to adopt an alternative behavior. A counterfactual explanation aims to
 1655 explain the agent’s decision-making process. Modifying the observation informs
 1656 the observer about the adjustments required for the agent to adopt a different
 1657 behavior. The family of methods available to generate counterfactuals is the *use*
 1658 *of the Latent Space (LS) for counterfactual generation*.

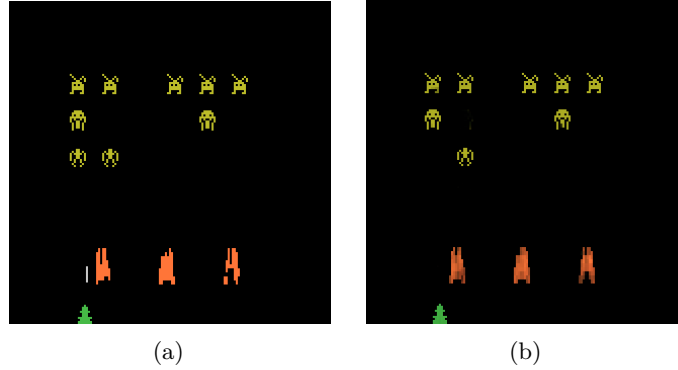


Figure 4.5: Counterfactual explanation of the agent’s behavior for the game *Space Invaders* [Olson et al., 2021]. (a) is the original observation, for which the agent selects the action “LEFT”, whereas (b) is the counterfactual observation, for which the agent selects the action “RIGHT_FIRE”.

Latent Space Usage for Counterfactual Generation

In this family of methods, **generative models** are used [Bond-Taylor et al., 2021] to exploit the Latent Representation (LR). This type of model learns to generate new data by imitating the training data distribution. In this context, generative models aim to produce alternative observations that are likely to alter the agent’s actions (see Figure 4.5) [Olson et al., 2021].

The main challenge of this approach lies in the fact that, to serve as explanations, counterfactuals must be both **close** and **feasible**. A close counterfactual refers to one that does not differ significantly from the original observation. A feasible counterfactual corresponds to an observation that is meaningful within the considered environment.

4.3.4 Agent Trajectory

Until now, we have presented methods that focus on explaining individual decisions made by the agent. We now consider methods that aim to provide insight into the agent’s strategy by analyzing its behavior over time.

The most direct approach consists in visualizing agent trajectories, most commonly through complete episodes. By analyzing these trajectories, the observer can infer the dynamics underlying the agent’s behavior and how its decisions evolve through time. This type of visualization is not specific to XRL. In practice, visualizing agent episodes is a standard step in the development and evaluation of RL. It provides a simple way to verify that the agent behaves as expected and to detect potential implementation errors.

This simple approach can be further enhanced using families of methods such as *hierarchical decomposition* and *extraction via critic analysis*.

Extraction via Critic Analysis

In this family of methods, the predictions of the critic are used to determine which **trajectories to present to the observer** [Sequeira and Gervasio, 2020].

A set of states is selected from multiple trajectories. For a given state, all possible actions are examined. The state is assigned a value equal to the difference between the critic’s prediction for the lowest-valued action and the prediction for the highest-valued action. Trajectories containing the states with the highest values according to this procedure are then chosen to be presented to the observer. These trajectories are considered **critical moments** of the agent and are therefore relevant for analysis. For this family of methods, the *selected trajectories* represent the interpretable information provided to the observer.

1694 *Hierarchical Decomposition*

Hierarchical agents are composed of a set of **sub-policies** specialized for specific tasks within subsets of the environment’s states. This family of methods distributes the optimization of the reward function across multiple models, significantly simplifying the learning process. Moreover, this **decomposition** of the policy function can also provide explainability in the agent’s strategy [Beyret et al., 2019]. If a sub-policy is selected, it indicates that the agent will execute a specialized behavior. This behavioral specialization provides insight into the strategy the agent employs to maximize the reward function. The agent’s *trajectory*, together with the *sub-policy* selected for each action, is used as interpretable information.

1705 4.3.5 Trajectory Prediction

Making predictions about the future of a trajectory allows one to form a representation of what the system expects, based on its training experience. These predictions are produced using value functions. Such functions provide insights into the future evolution of the agent’s trajectory. By their nature, this form of explanation informs the observer about the strategy adopted by the agent. Value functions provide an estimate of the expected cumulative reward from a given state. Consequently, a value function maps an observation to a scalar value in $\mathbb{R}$.

This aggregated prediction provides limited information about the underlying factors that contribute to this value. It does not directly reveal which aspects of the task are being optimized or how the predicted return is composed. To improve the interpretability of value-based explanations, a family of methods known as *critic enrichment* can be employed.

1719 *Critic Enrichment*

In order to obtain interpretable information from the predictions of the **critic**, it is possible to **decompose** the critic into multiple **sub-functions** [Juozapaitis et al., 2019]. The reward function is thus decomposed into multiple sub-reward functions, each representing a **sub-objective** of the task to be accomplished. Similarly, for each of these sub-reward functions, corresponding sub-critic functions are defined, along with a function that combines their outputs. During training, the agent selects its actions so as to maximize this combination function of the sub-critics. Using this approach, for each observation and action, a prediction is obtained regarding the sub-objectives the agent is attempting to maximize (see Figure 4.6). This *prediction* is then used as the explanation for the observer.

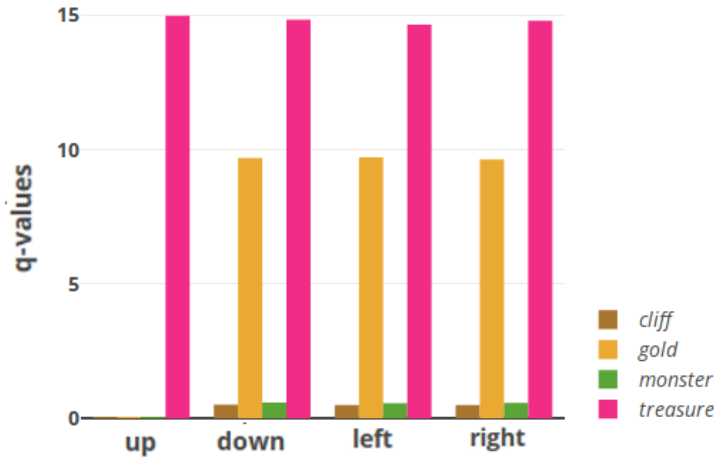


Figure 4.6: Prediction of sub-rewards as a function of the action selected by the agent for a given state [Juozaipaitis et al., 2019]. Depending on the action chosen by the agent, one can determine which sub-reward it aims to maximize.

4.3.6 Interaction Model

Modeling the interactions between an agent and its environment makes it possible to understand how the agent’s decisions are made and what impact they have on the environment. By incorporating these interactions into an explanatory model, one can better grasp the complex relationships between the agent and its environment, thereby identifying the determining factors in the decision-making process. By explicitly handling the agent–environment interaction, this type of explanation aims to explain the agent’s strategy. The families of methods used to obtain explanations in the form of agent–environment interactions are: *Bayesian network learning* and *LS usage for MDP generation*.

Bayesian Network Learning

The evolution of an agent within an environment can be observed as a set of **random variables** connected through a **Bayesian network** [Madumal et al., 2020]. Random variables are defined to represent both external aspects (the environment) and internal aspects of the agent (the actions). By analyzing trajectories, it becomes possible to infer causal relationships among these random variables. At the end of this process, a weighted causal graph is obtained, which serves as an explanation for the observer (see Figure 4.7). This *Bayesian network*, representing the interactions between the agent and the environment, is then used as the explanation.

Latent Space Usage for MDP Generation

Each layer of a NN can be viewed as a projection from one space to another [Zahavy et al., 2017]. As information propagates through successive layers of a NN, it becomes increasingly abstract. It can therefore be assumed that two vectors that are close in this projected space are also close in terms of the agent’s

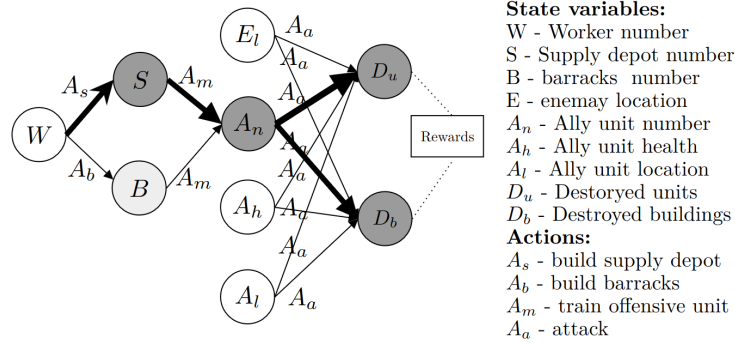


Figure 4.7: Agent-environment interaction model for the video game *StarCraft II* [Madumal et al., 2020]. Nodes represent random variables, and edges represent causal relationships between them.

perception. Based on this observation, it is possible to **redefine an MDP** by leveraging the **abstract representations** produced by the final layers of a NN. Once the new MDP has been constructed, a model of the interaction between the agent and its environment is obtained. This model, represented as a *graph*, is then used as the explanation.

Figure 4.8 illustrates this idea using a two-dimensional *t-SNE* projection of the LS of the NN used to represent the policy. Each point is colored according to the value function’s prediction for the corresponding state. An agreement can be observed between the structure of this projected space and the critic’s predictions: states that are assigned similar values by the critic also tend to be located close to one another in the projection. This agreement makes it possible to manually identify clusters of states, which likely correspond to states that are perceived as similar by the agent.

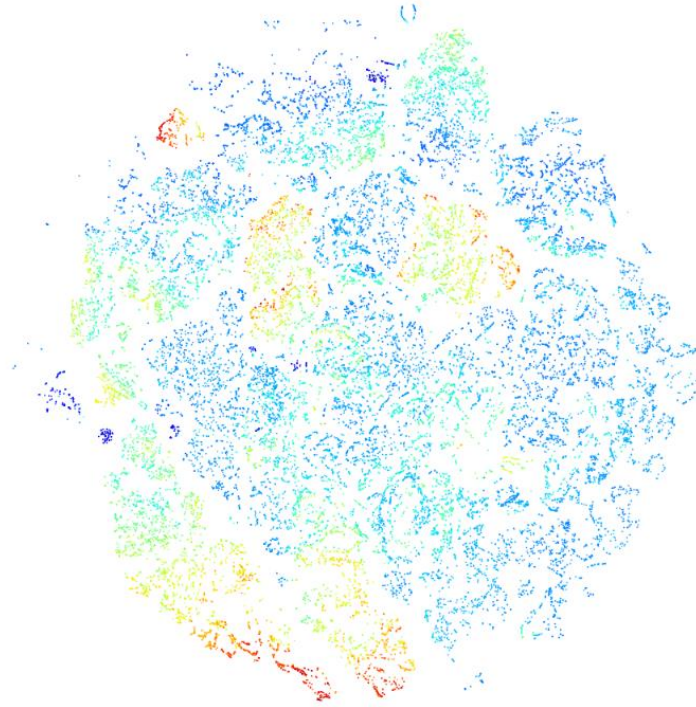


Figure 4.8: t -SNE projection of a neural network's latent space [Zahavy et al., 2017]. Each point corresponds to a state, and its color represents the value function's prediction for that state, with warmer colors indicating higher predicted values.

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